大量科技
TALIANG TECHNOLOGY

SMT - SECS/GEM

User manual

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大量科技股份有限公司 *TALIANG TECHNOLOGY*
桃園市八德區友聯街 **49** 號 *No. 49, Youlian St., Bade Dist., Taoyuan City 334009, Taiwan*
TEL : (03)3686368 / FAX : (03)3685811

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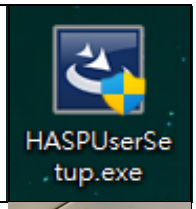
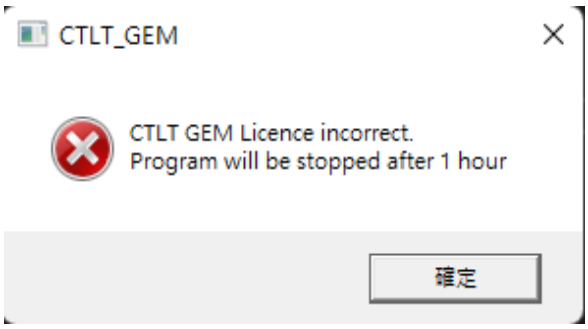
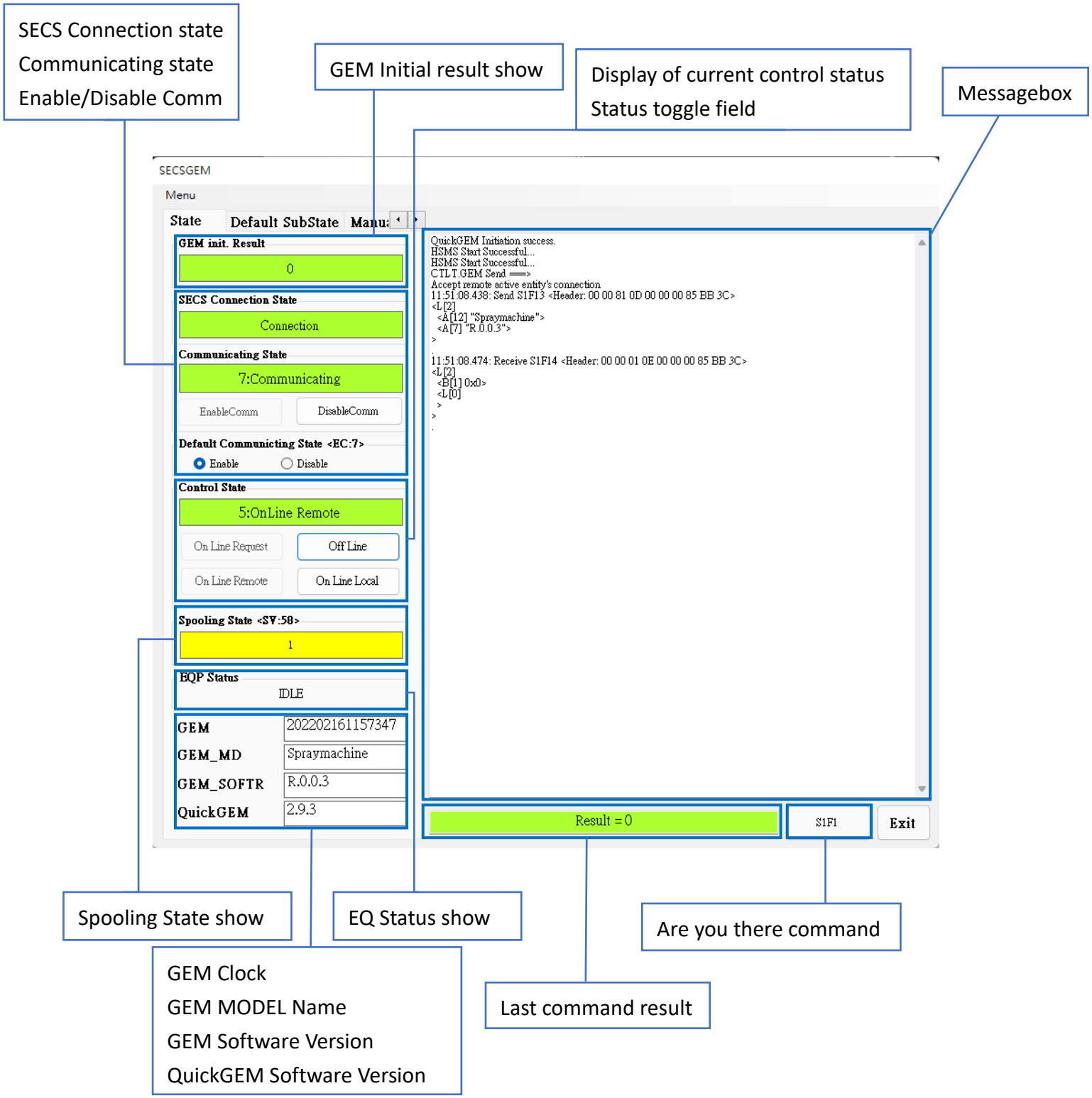
SECS/GEM Driver & Dongle Install

STEP1 : **Install HASPUserSetup.exe**	
STEP2 : **Insert SECS/GEM Dongle to Computer's USB port**	

※If the installation fails or the USB cannot be read, the message box will pop up as shown below.



SECS/GEM Interface - State



EnableComm：Start Communication

Part	Description
CGWrapper	CTLT.GEM Control

Returns：
0 ： Successful
10 ： CTLT.GEM尚未執行Communication功能初始化(本項異常不會記錄在Log)

DisableComm：Stop Communication

Part	Description
CGWrapper	CTLT.GEM Control

Returns：
0 ： Successful
10 ： CTLT.GEM尚未執行Communication功能初始化(本項異常不會記錄在Log)

On Line Request：Send the request to connect with the remote On Line

Part	Description
CGWrapper	CTLT.GEM Control

Returns：
0 ： Successful
1 ： 目前Control State不是Equipment Off Line State，使用者禁止On Line(本項異常不會記錄於Log)

Off Line：Send the request to disconnect with the remote On Line

Part	Description
CGWrapper	CTLT.GEM Control

Returns：
0 ： Successful
1 ： 目前Control State不是On Line State或Host Off Line State，使用者禁止Off Line(本項異常不會記錄於Log)

On Line Remote：Switch to remote control status

Part	Description
CGWrapper	CTLT.GEM Control

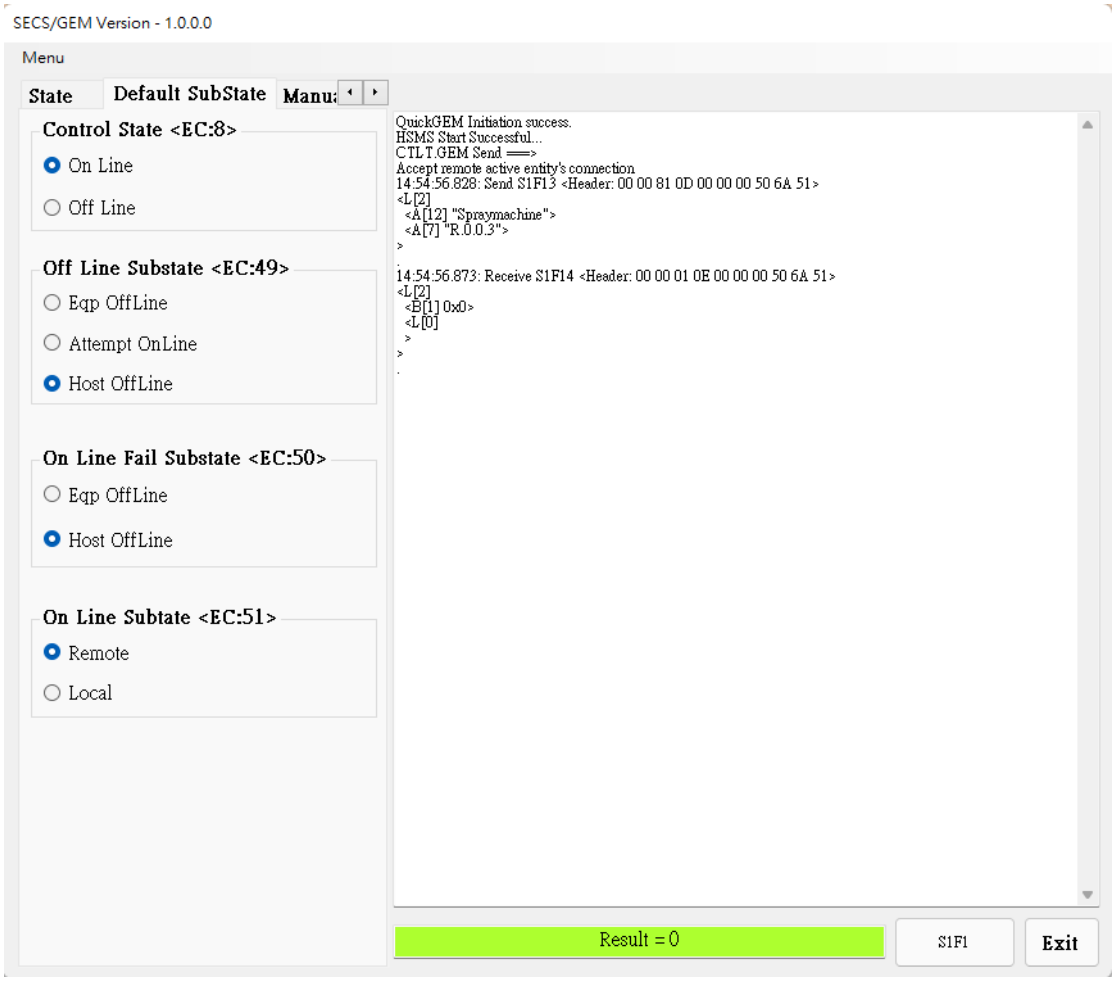
Returns：
0 ： Successful
1 ： 目前Control State不是On Line Local State，故使用者不能要求On Remote Local(本項異常不會記錄於Log)

On Line Local：Switch to local control status

Part	Description
CGWrapper	CTLT.GEM Control

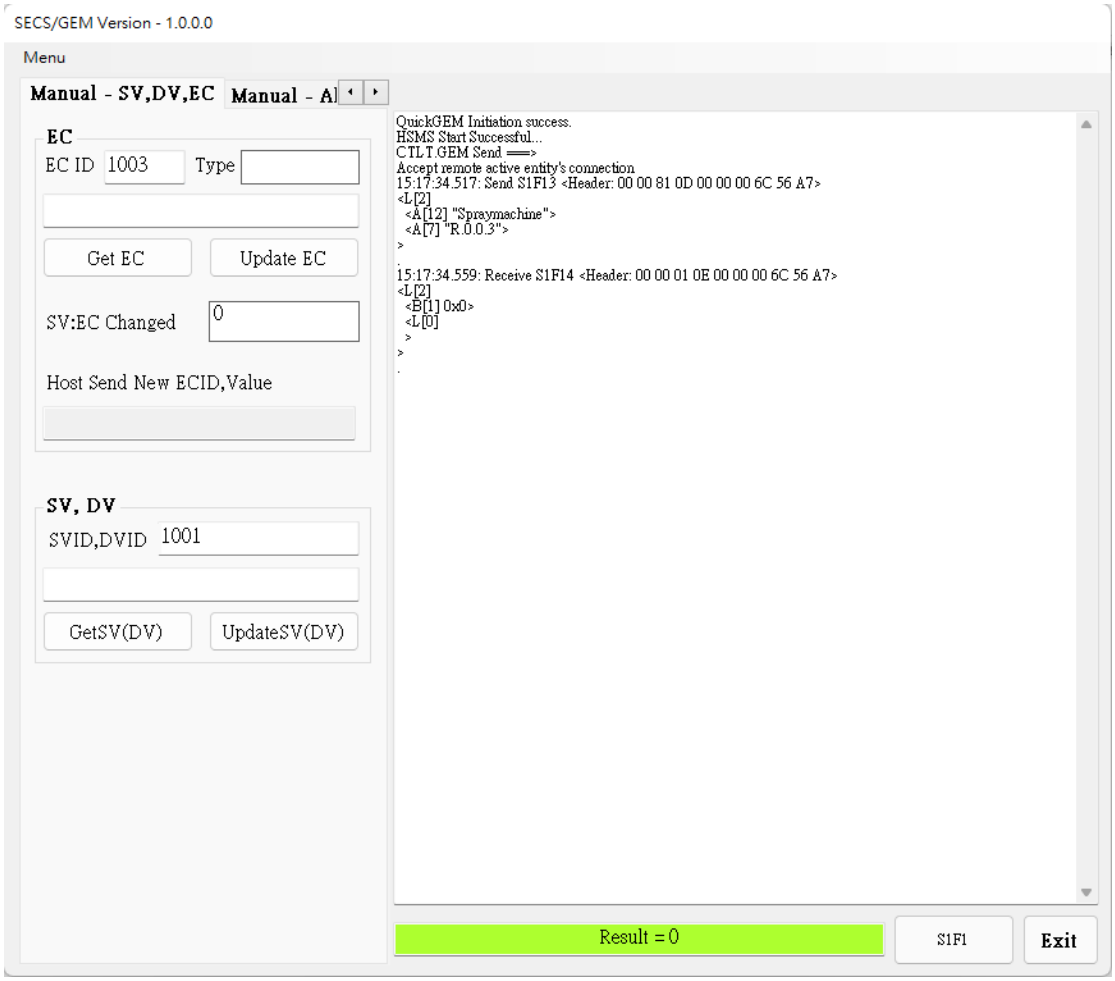
Returns：
0 ： Successful
1 ： 目前Control State不是On Line Remote State，故使用者不能要求On Line Local(本項異常不會記錄於Log)

SECS/GEM Interface – Default Substate



Switch EC:8, EC:49, EC:50, EC:51 ‘s sub state, See [System EC page (p.23)] for details

SECS/GEM Interface – Manual SV,DV,EC



Update EC : Update specified Equipment Constant(EC) value

Part	Description
CGWrapper	CTLT.GEM Control
ECID	[in] Equipment Constant Identifier
Value	[in]Assign value of each equipment constant. The type of value is Variant which means that the value type is assigned depend on Format of ECID defined in initial file.

Returns :
0 : Successful
10 : CTLT.GEM尚未初始化(本項異常不會記錄在Log)
901 : Value之型態無法於COM介面轉換(本項異常不會記錄在Log)

Update SV(DV)：Update specified Status Variable(SV) and Data Value(DV) value

Part	Description
CGWrapper	CTLT.GEM Control
SVID/DVID	[in]Status Variable Identifier / Data Value Identifier
Value	[in]Assign value of each status variable. The type of value is Variant which means that the value type is assigned depend on Format of SVID/DVID defined in initial file.

Returns：
0 ： Successful
10 ： CTLT.GEM尚未初始化(本項異常不會記錄在Log)
901： Value之型態無法於COM介面轉換(本項異常不會記錄在Log)

GetEC：Load specified Equipment Constant value

Part	Description
CGWrapper	CTLT.GEM Control
SVID/DVID	[in]Equipment Constant Identifier
Format	[out]Store the format of each ECID. The format is defined in EC_DATA_TYPE as following： _BINARY_TYPE (33): Binary. _BOOLEAN_TYPE (37): Boolean. _ASCII_TYPE (65): ASCII String. _INT1_TYPE (101): 1 byte integer. _INT2_TYPE (105): 2 byte integer. _INT4_TYPE (113): 4 byte integer. _UINT1_TYPE (165): 1 byte unsigned integer. _UINT2_TYPE (169): 2 byte unsigned integer. _UINT4_TYPE (177): 4 byte unsigned integer. _FT4_TYPE (145): 4 byte floating number. _FT8_TYPE (129): 8 byte floating number.
Value	[out] Store the value of specific ECID.

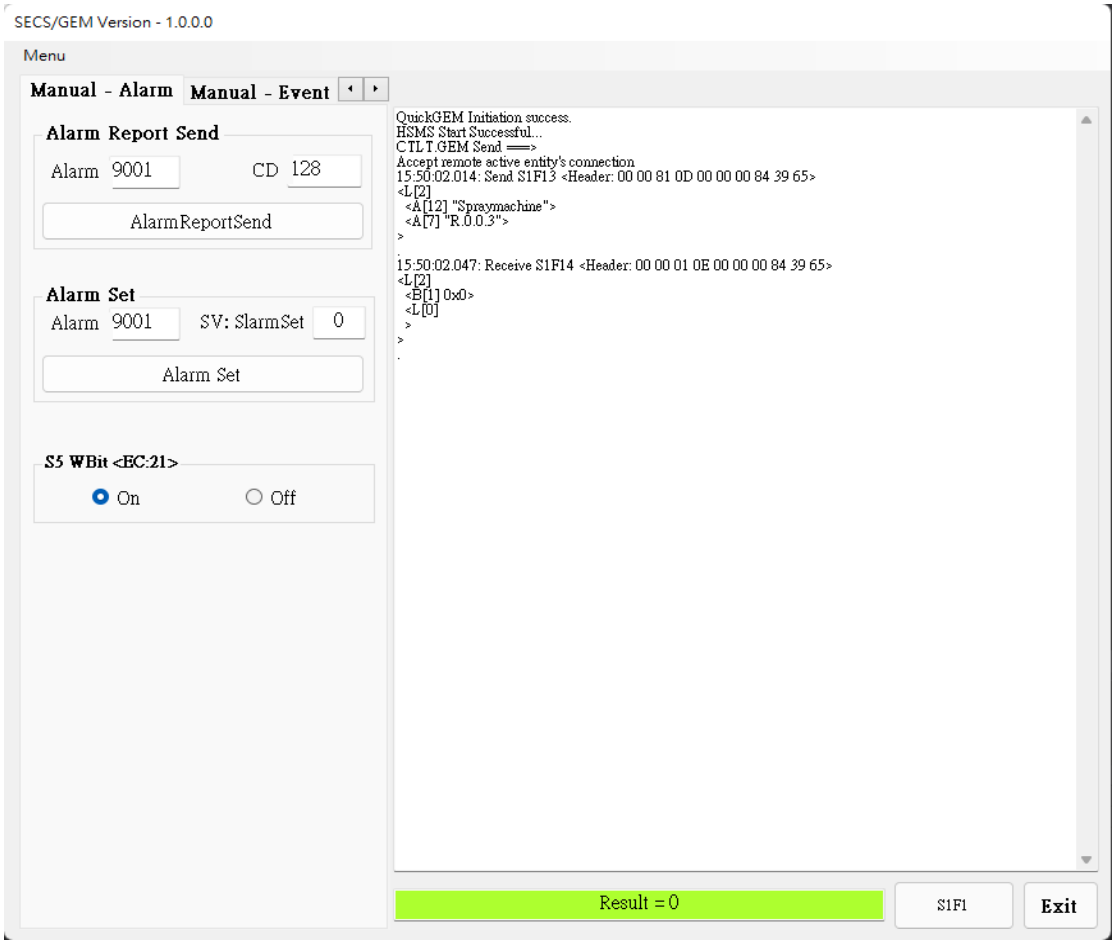
Returns：
0 ： Successful
10 ： CTLT.GEM尚未初始化(本項異常不會記錄在Log)

GetSV(DV)：Load specified Status Variable(SV) and Data Value(DV) value

Part	Description
CGWrapper	CTLT.GEM Control
SVID/DVID	[in]Status Variable Identifier / Data Value Identifier
Format	[out]Store the format of each SVID. The format is defined in SV_DATA_TYPE as following： _BINARY_TYPE (33): Binary. _BOOLEAN_TYPE (37): Boolean. _ASCII_TYPE (65): ASCII String. _INT1_TYPE (101): 1 byte integer. _INT2_TYPE (105): 2 byte integer. _INT4_TYPE (113): 4 byte integer. _UINT1_TYPE (165): 1 byte unsigned integer. _UINT2_TYPE (169): 2 byte unsigned integer. _UINT4_TYPE (177): 4 byte unsigned integer. _FT4_TYPE (145): 4 byte floating number. _FT8_TYPE (129): 8 byte floating number.
Value	[out] Store the value of specific SVID/DVID.

Returns：
0 ： Successful
10 ： CTLT.GEM尚未初始化(本項異常不會記錄在Log)

SECS/GEM Interface – Alarm



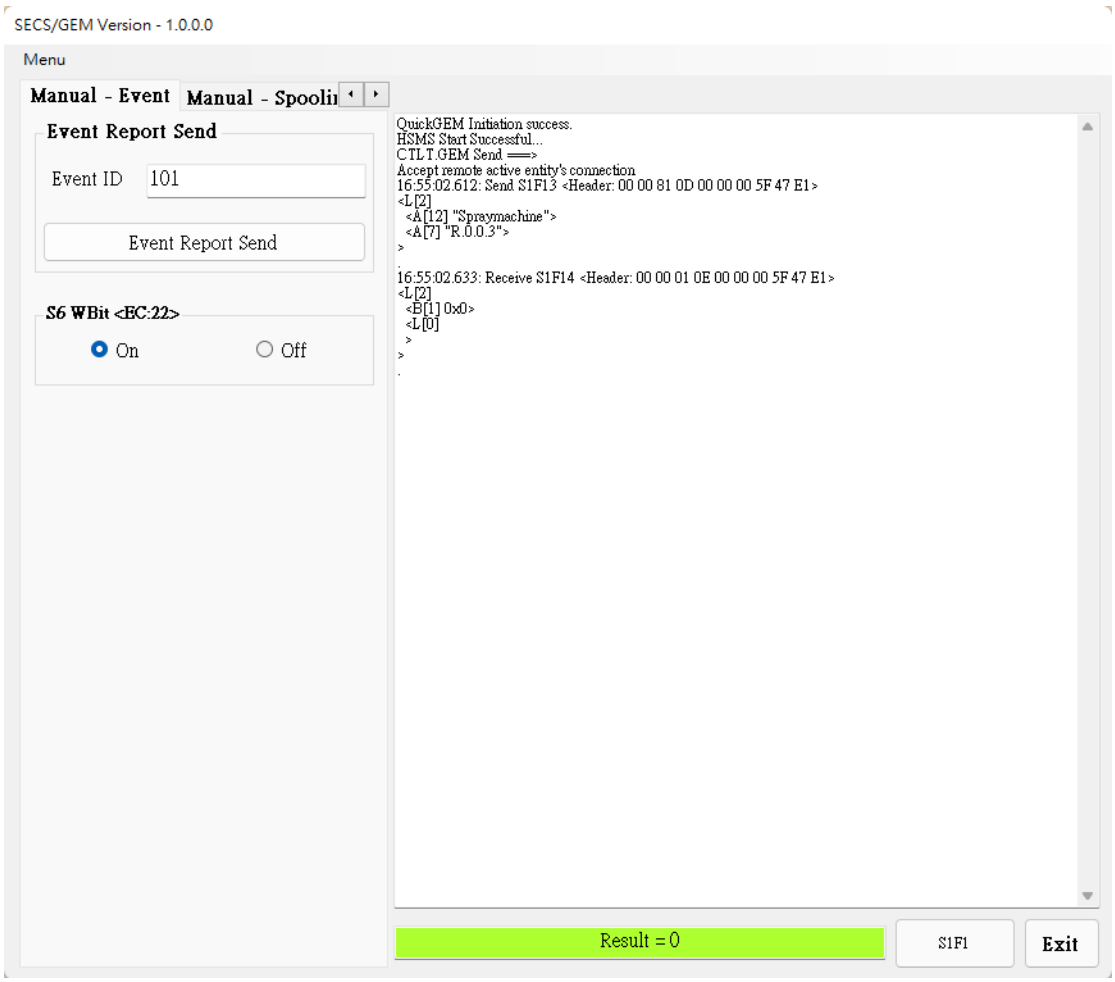
AlarmReportSend：Based on specified Alarm ID send alarm message

Part	Description
CGWrapper	CTLT.GEM Control
AlarmID	[in]Specifies Alarm ID to be sent.
AlarmCD	[in]Specifies Alarm Code to be sent. 依照E30規範AlarmCD，只能使用0以及128(CTLT.GEM並未管制AlarmCD) 0：表示Alarm Reset或結束 128：表示發生Alarm

Returns：
0 ： Successful(訊息成功送出或被Spooled)
1 ： 訊息太大超過CTLT.GEM Buffer容量，故無法送出
3 ： 訊息無法送出，因系統處於Control Off Line State，故禁止送出訊息
4 ： 訊息無法送出，因CLTL.GEM尚未進入Communicating State，故Message禁止送出
5 ： 系統已進入Spool Active State，而S6F11不在Enable Spool List名單內，故不會被Spooled
(以上異常皆不會記錄在Log)

S5 WBit <EC:21>：See [System EC page (p.23)] for details

SECS/GEM Interface – Event



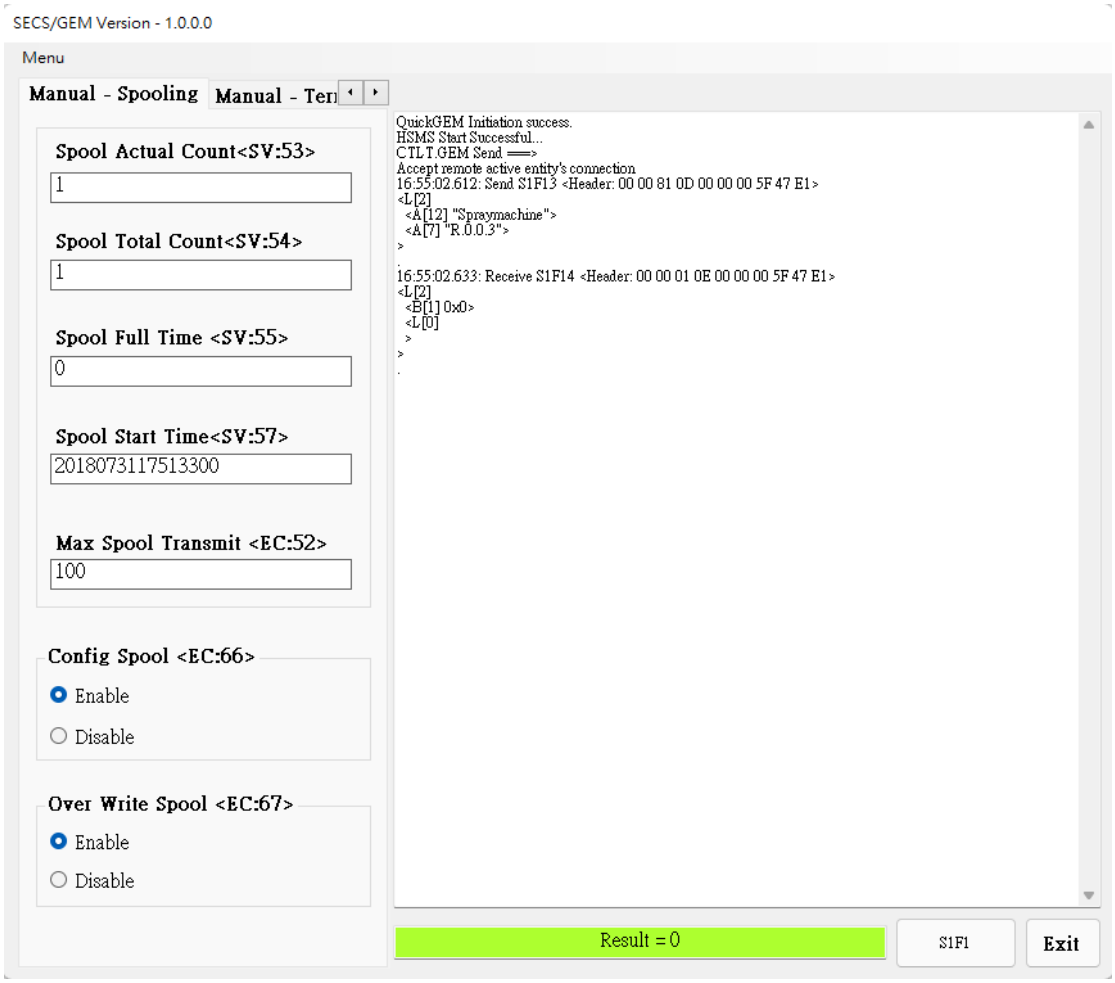
EventReportSend：Based on specified Event ID send event message

Part	Description
CGWrapper	CTLT.GEM Control
EventID	[in]Specifies Event ID to be sent

Returns：
0 ： Successful
1 ： Event ID找不到，無法送出
3 ： 訊息無法送出，因系統處於Control Off Line State，故禁止送出訊息
4 ： 訊息無法送出，因CLTL.GEM尚未進入Communicating State，故Message禁止送出
5 ： 系統已進入Spool Active State，而S6F11不在Enable Spool List名單內，故不會被Spooled
9 ： 該Event ID 被Disable，故無法送出
10 ： 訊息太大超過CTLT.GEM Buffer容量，故無法送出
(以上異常皆不會記錄在Log)

S6 WBit <EC:22>：See [System EC page (p.23)] for details

SECS/GEM Interface – Spooling

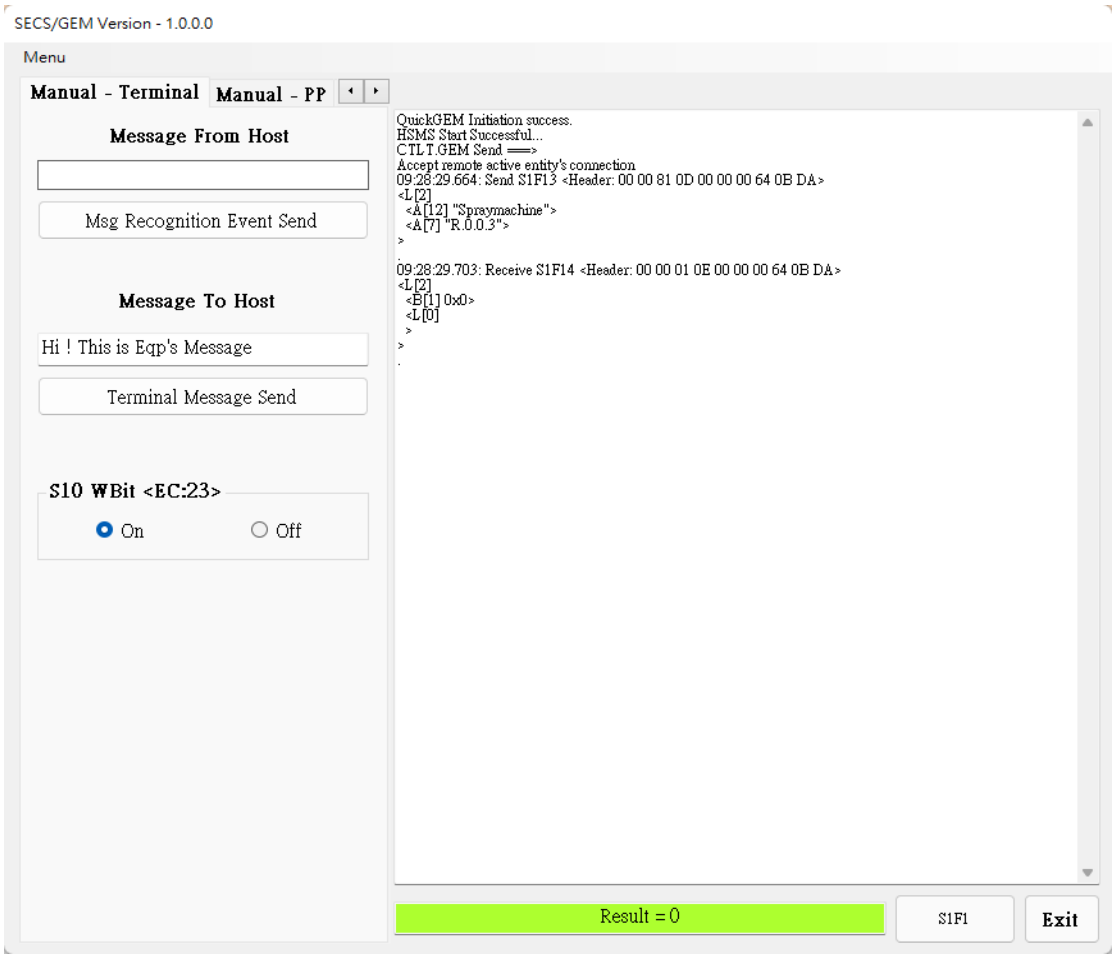


Display SV:53, SV:54, SV:55, SV:57 current value, See [System SV page (p.20)] for details

Display EC:52current value, See [System EC page (p.23)] for details

Open/Close EC66, EC67 enable state, See [System EC page (p.23)] for details

SECS/GEM Interface – Terminal



Msg Recognition Event Send: Send terminal message

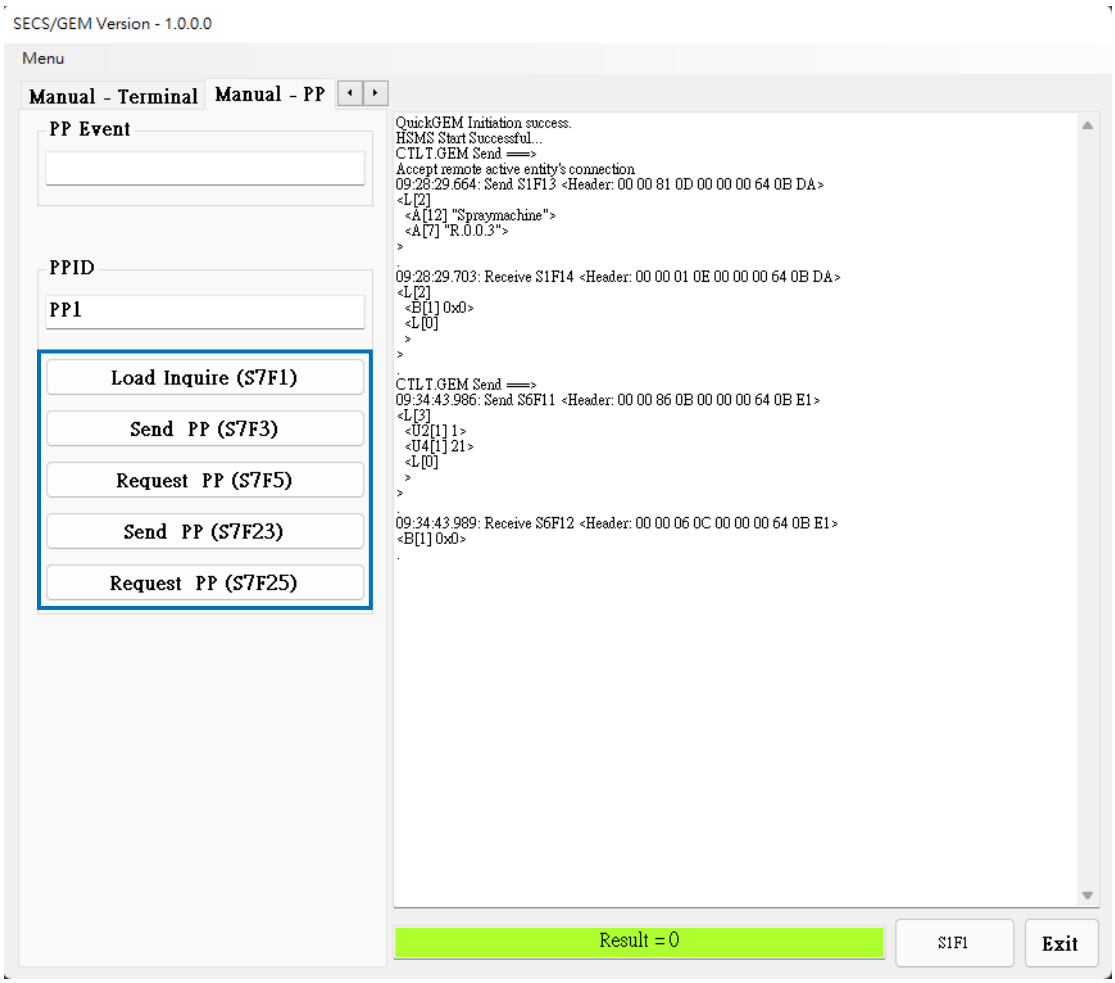
Terminal Message Send: Send terminal message

Part	Description
CGWrapper	CTLT.GEM Control
Message	[in]Specifies Message Text to be sent

Returns :
0 : Successful
10 : CTLT.GEM尚未初始化(本項異常不會記錄在Log)

S10 WBit <EC:23> : See [System EC page (p.23)] for details

SECS/GEM Interface – PP

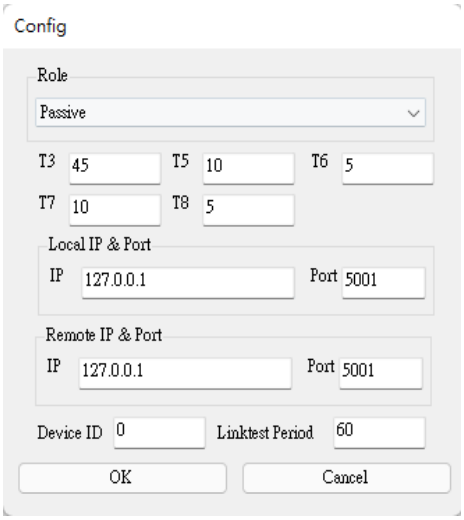


Function specified by excute CmdID

Part	Description
CGWrapper	CTLT.GEM Control
CmdID	[in]命令代號： CMD_PP_LOAD_INQUIRE(11)： Send process program load inquire(S7F1) CMD_UMFORMATTED_PP_SEND(12)： Send process program send(S7F3) CMD_UNFORMATTED_PP_REQUEST(13)： Send process program request(S7F5) CMD_FORMATTED_PP_SEND(14)： Send formatted process program send(S7F23) CMD_FORMATTED_PP_REQUEST(15)： Send formatted process program request(S7F25)
Param1	[in]process program ID(PPID)
Param2	[in]Null

Returns：
1 ： 本機找不到Param1所指定的PPID，無法送出Process Program(for S7F1、S7F3、S7F23)
3 ： 訊息無法送出，因系統處於Control Off Line State，故禁止送出訊息
4 ： 訊息無法送出，因CLTL.GEM尚未進入Communicating State，故Message禁止送出
5 ： 系統已進入Spool Active State，而S6F11不在Enable Spool List名單內，故不會被Spooled
6 ： 本機已存在Param1所指定的PPID且PPOverwriteEnable = No，所以無法送出PP Request訊息(S7F5、S7F25)以要求Host送出Process Program
7 ： CmdID錯誤
10 ： CTLT.GEM尚未執行初始化(本項異常不會記錄在Log)
901： Value型態無法於COM介面轉換(此異常不會記錄在Log)

Config Interface



Role : Set the machine as the active side (Host) or the passive side (Equipment)

T3 : Set the time for the sender to wait for the receiver to return a message after sending the data message

T5 : After the TCP/IP connection is set to fail, the active side needs to wait for T5 time before reconnecting

T6 : Set the time for the sender to wait for the receiver to return a message after sending a control message

T7 : Set the time to wait to enter Selected Stats after entering Connected Stats

T8 : Set the maximum time interval allowed between characters in the message when the receiver receives the message

Local IP & Port : Set local IP & Port

Remote IP & Port : Set remote IP & Port

Device ID : Set Device ID

Linktest Period : Set the interval for sending test messages periodically (limited by T6)

System SV

SVID	Type	Length	SV Name	Remark	說明
1	ASCII	39	License Code		CTLT.GEM Licence的文字說明
2	UINT_1		License Status	0 : No License 1 : Normal License 2 : Small License	CTLT.GEM 授權狀態代碼
3	ASCII	19	GemClock		CTLT.GEM系統時間，格式依照EC : 68 (GemTimeFormat)制定
4	UINT_1		GemControlState	1 = Offline / Equipment Offline 2 = Offline / Attempt Online 3 = Offline / Host Offline 4 = Online Local 5 = Online Remote	目前的Control State
5	UINT_1		GemLinkState	0 : Disable 1 : Enable / Not Communicating 2 : Enable / Communicating	Communicating State
6	UINT_1		SECSCommunicationMode	0 : HSMS Mode(Default) 1 : SECS Mode(Set by APP)	HSMS/SECS-I通訊模式設定，本項預設值為HSMS Mode，若APP使用SECS-I Mode，務必要將本項設定為1，CTLT.GEM才能正常處理Communicating State Model
11	UINT_4		GemPreviousCEID		本項紀錄最後一次所送出的Event ID編號
12	UINT_1		GemOffLineSubState	1 : Eqp Off-Line 2 : Attempt On-Line 3 : Host Off-Line Others : Not at Off-Line	Off-Line的子狀態
13	UINT_1		GemPreviousControlState	1 : Eqp Off-Line 2 : On-Line Local 3 : On-Line Remote	前一次Control State的狀態
14	UINT_1		GemPreviousProcessState	Set by APP	機台前一次的Process State狀態，本項由APP進行Update(本項可刪除)
15	UINT_1		GemProcessState	Set by APP	機台目前的Process State狀態，本項由APP進行Update(本項可刪除)
24	ASCII	6	GemMDLN	Set by APP	機台型號，本項務必使用APP Update

25	ASCII	6	GemSOFTREV	Set by APP	機台軟體，本項務必使用APP Update(本項應以Hard Code方式設計)
39	LIST		AlarmsEnabled		Alarm為Enabled的List資料
40	LIST		AlarmSet		Alarm為Set的List資料
41	LIST		EventsEnabled		Event為Enabled的List資料
42	ASCII	256	GemPPExecName	Set by APP	目前機台所選擇要執行的Process Program Name，本項由APP執行Update(本項可刪除)
43	UINT_1		PPFormate	1：僅支援Unformatted Process Programs 2：僅支援Formatted Process Programs 3：同時支援Unformatted Process Programs和Formatted Process Programs	提供目前的Process Program支援Formatted或Unformatted的功能
53	UINT_4		GemSpoolCountActual		本項記錄目前Spool訊息的實際數量
54	UINT_4		GemSpoolCountTotal		本項紀錄從Spool Active之後，所有被Spool的訊息的數量
55	ASCII	19	GemSpoolFullTime		本項紀錄從Spool進入Full State的時間點
57	ASCII	19	GemSpoolStateTime		本項記錄從Spool進入Active State的時間點
58	UINT_1		GemSpoolState	1：Inactive 2：Active	目前Spool的主要狀態
80	ASCII	20	GemSoftWareRevision		本項為取得CTLT.GEM的軟體版本(Ex：” X.X.X”)

System DV

SVID	Type	Length	SV Name	Remark	說明
9	ASCII	80	GemPPChangeName	Set by APP	當APP有新增、更改、刪除Process Program時，務必要更新本項，以供Host讀取有異動的Process Program名稱
10	UINT_1		GemPPChangeStatus	0 : Created 1 : Changed 2 : Deleted Set by APP	當APP有新增、更改、刪除Process Program時，務必要更新本項，以供Host讀取異動的原因
38	ASCII	19	Alarm ID		最後一次Alarm Report的Alarm ID
46	UINT_1		GemECIDChanged		本項紀錄最後一次被UpdateEC的ECID
47	UINT_1		GemECValueChanged		本項紀錄最後一次被UpdateEC的新EC值
48	UINT_1		GemPreviousECValue		本項紀錄最後一次被UpdateEC的原EC值

System EC

SVID	Type	Min	Max	Default	Unit	EC Name	Remark	說明
7	UINT_1	0	1	0		GemInitCommState	0：Disable 1：Enable	控制Communicating State Model的Default Entry State
8	UINT_1	1	2	1		GemInitControlState	1：Off-Line 2：On-Line	為控制Control State Model的Default Entry State
21	ASCII	0	1	1		GemWBitS5	0：Not Set 1：Set	控制於傳送S5F1時所帶的Wait Bit值
22	UINT_1	0	1	1		GemWBitS6	0：Not Set 1：Set	控制於傳送S6F1時所帶的Wait Bit值
23	UINT_1	0	1	1		GemWBitS10	0：Not Set 1：Set (For S10F1)	控制於傳送S10F1時所帶的Wait Bit值
44	UINT_1	1	10000	5	Sec	GemEtstabCommDelay		控制Communicating State Model 進入Wait Delay Substate的Time Out時間
49	UINT_4	1	3	1		GemOffLineSubState	1：Eqp. Off-Line 2：Attempt On-Line 3：Host Off-Line	控制Control State Model於Off-Line Substate
50	UINT_1	1	3	1		GemOffLineSubState	1：Eqp. Off-Line 3：Host Off-Line	控制Control State Model的Attemp On-Line無法On-Line時要進入的Off-Line Substate
51	UINT_1	4	5	4		GemPreviousControlState	4：On-Line / Local 5：On-Line / Remote	控制Control State Model於On-Line Substate的Default Entry State
52	UINT_1	1	2E+08	100		GemMaxSpoolTransmit		控制Spool State Model的Transmit Spool SubState一次最多可以傳送的訊息數量。當送達此數量，並且Spool尚有未傳送的訊息時，則Spool State Model會回到NO SPOOL OUTPUT Substate

66	UINT_1	0	1	0		GemConfigSpool	0：Disable 1：Enable	控制Spooling功能是否要啟動。 注意：依照SEMI E30的規範，若系統已進入Spooling Active State後再Disable Spooling的功能，並不會改變Spooling目前於SPOOL ACTIVE STATE的狀態
67	ASCII	0	1	0		GemOverWriteSpool	0：Do not overwrite 1：Overwrite	當Spool的空間滿了之後，若有新的訊息要加入到Spool時，是否能夠取代Spool中舊有的訊息。 注意：若設定Do not overwrite，新訊息將會被捨棄
68	ASCII	0	3	1		GemTimeFormat	0：12-bytes 1：16-bytes 2：14-bytes 3：ISO8601 format	設定”時間”的傳輸格式
71	LIST	1	6	5		GemDATAIDFormat	1：INT_1 2：INT_2 3：INT_4 4：UINT_1 5：UINT_2 6：UINT_4	送S6F11時，設定DATAID之格式。本項為CTLT.GEM自動產生的流水號值，範圍於0~65535之間
75	LIST	1	6	5		GemSAMPLEFormat	1：INT_1 2：INT_2 3：INT_4 4：UINT_1 5：UINT_2 6：UINT_4	Trace data功能中，送S6F1訊息時，設定訊息SAMPLN item的格式

System Event

SVID	Event Name	說明
3	GemProcessProgramChange (Set by APP)	當機台Process State有變動時，必須要報Event給Host。本Event由APP負責送出，建議APP應該用獨立不同的Event告知Host目前的Process State (本項可刪除)
8	GemControlStateLocal	當GEM的Control State Model進入On-Line Local Substate時，核心會送出本訊息
9	GemControlStateRemote	當GEM的Control State Model進入On-Line Remote Substate時，核心會送出本訊息
20	GemEqConstChanged	當APP Call UpdateEC函式時(使用者修改EC)，核心會送出本訊息 註：當Host以S2F15改變設備EC時，核心並不會送出本訊息
21	GemMessageRecognition (Set by APP)	當APP收到S10F3時，Operator應該按下「訊息確認」的按鍵，送出確認的Event，已告知Host，Operator已閱讀訊息，本Event由APP負責送出。若Host送出多個S10F3，Operator只需按一次確認鍵，表示已讀取所有訊息
22	GemEqpOffLine	當GEM的Control State Model進Off-Line State時，核心會送出本訊息
23	GemSpoolingActived	當GEM的Spool State Model進入SPOOL ACTIVE State時，核心會產生本訊息並且存到Spool中。因此GemSpoolingActived的Event一定會是第一個被Spool的訊息
24	GemSpoolingDeactived	當GEM Spool中的訊息被送完或被清除時，核心會送出本訊息
25	GemSpoolTransmitFailure	當GEM於TRANSMIT SPOOL Substate傳送訊息時，若發生Communicating State Model進入NOT COMMUNICATING State則於重新進入COMMUNICATING State時，核心會送出本訊息 注意：本Event非SEMI E30規範中必須送的Event，可以設定本Event為Disable，但請勿刪除本項Event

Equpiment SV

ID	TYPE	Lenght	NAME	Definition
1001	ASCII	1000	MagazineIDCheck	入料端 MagazineID 確認
1002	ASCII	1000	BoatBarcodeCheck	入料端 BoatID 確認
1003	ASCII	1000	Pass1MagazineIDCheck	出料端 Pass1MagazineID 確認
1004	ASCII	1000	Pass2MagazineIDCheck	出料端 Pass2MagazineID 確認
1005	ASCII	1000	NGMagazineIDCheck	出料端 NGMagazineID 確認
1006	ASCII	1000	EmptyMagazineIDCheck	出料端 EmptyMagazineID 確認
1007	ASCII	1000	BoatAssingNGBoatBarcodeCheck	出料端 空 Boat 回流登記 NGBoat 確認
1008	ASCII	1000	UILockCommand	遠端切換 UI 鎖定/解鎖狀態指令
1009	ASCII	1000	EquipmentOperationalCommand	遠端切換設備啟動/停止狀態指令
1010	ASCII	1000	EquipmentEndLotCommand	遠端控制設備結批指令
1011	ASCII	1000	RemoteActionResult	遠端指令執行結果
1012	ASCII	1000	EmptyBoatReleaseCheck	空 Boat Release 狀態 確認
1013	ASCII	1000	EquipmentStartRunCheck	設備啟動狀態確認

Equpiment DV

ID	TYPE	Lenght	NAME	Definition
4001	ASCII	1000	LotID	Lot ID
4002	ASCII	1000	MagazineID	入料端 MagazineID
4003	ASCII	1000	Pass1MagazineID	出料端 Pass1MagazineID
4004	ASCII	1000	Pass2MagazineID	出料端 Pass2MagazineID
4005	ASCII	1000	NGMagazineID	出料端 NGMagazineID
4006	ASCII	1000	EmptyMagazineID	出料端 EmptyMagazineID
8001	ASCII	1000	BoatBarcode	入料端 Boat 條碼
8002	ASCII	1000	PassBoatBarcode	出料端 PassBoat 條碼
8003	ASCII	1000	NGBoatBarcode	出料端 NGBoat 條碼
8004	ASCII	1000	EmptyBoatBarcode	出料端 空 Boat 條碼
8005	ASCII	1000	BoatAssingNGBoatBarcode	出料端 空 Boat 回流登記 NGBoat 條碼
8006	ASCII	1000	RecipeName	配方名稱
8007	ASCII	1000	Pass1MagazineUnitCount	出料端 Pass1MagazineID Unit 總數
8008	ASCII	1000	Pass2MagazineUnitCount	出料端 Pass2MagazineID Unit 總數
8009	ASCII	1000	NGMagazineUnitCount	出料端 NGMagazineID Unit 總數

Equpiment Event

ID	NAME	Definition
101	MagazineMoveInEvent	入料端 Magazine MoveIn 事件
102	BoatMoveInEvent	入料端 Boat MoveIn 事件
103	PassBoatMoveOutEvent	出料端 PassBoat MoveOut 事件
104	NGBoatMoveOutEvent	出料端 NGBoat MoveOut 事件
105	EmptyBoatMoveOutEvent	出料端 EmptyBoat MoveOut 事件
106	Pass1MagazineMoveInEvent	出料端 Pass1Magazine MoveIn 事件
107	Pass2MagazineMoveInEvent	出料端 Pass2Magazine MoveIn 事件
108	NGMagazineMoveInEvent	出料端 NGMagazine MoveIn 事件
109	EmptyMagazineMoveInEvent	出料端 EmptyMagazine MoveIn 事件
110	Pass1MagazineMoveOutEvent	出料端 Pass1Magazine MoveOut 事件
111	Pass2MagazineMoveOutEvent	出料端 Pass2Magazine MoveOut 事件
112	NGMagazineMoveOutEvent	出料端 NGMagazine MoveOut 事件
113	BoatAssingNGBoatEvent	出料端 空 Boat 回流登記 NGBoat 事件
114	RemoteActionResultEvent	遠端指令執行結果事件
115	EquipmentStartRunEvent	設備啟動狀態事件

Equpiment Alarm - Loader

ID	Name	Definition
10001	Loader - Rail door inspection abnormal	Loader - 門檢異常
10002	Loader - The emergency stop was pressed	Loader - 緊急停止被按下
10003	Loader - Air Pressure Low(X02)	Loader - 氣壓不足(X02)
10004	Loader - Jaws cylinder abnormal (X03)	Loader - 夾爪氣缸動檢(X03)-異常
10005	Loader - Jaws cylinder abnormal (X04)	Loader - 夾爪氣缸原點檢(X04)-異常
10006	Loader - Jaws Right side abnormal (X05)	Loader - 料匣感知-R(X05)-異常
10007	Loader - Jaws Left side abnormal (X06)	Loader - 料匣感知-L(X06)-異常
10008	Loader - Inlet area Z position wrong(X07)	Loader - 入料區-夾匣位置檢(X07)-異常
10009	Loader - Outlet area Z position wrong(X08)	Loader - 出料區-夾匣位置檢(X08)-異常
10010	Loader - Boat stuck(X09)	Loader - Boat 盤推移防撞檢(X09)-異常
10011	Loader - Inlet area arrive sensor error(X10)	Loader - 入料區-料匣到位檢(X10)-異常
10012	Loader - Inlet CV motor error(X11)	Loader - 入料區-料匣皮帶馬達(X11)-異常
10013	Loader - Outlet area arrive sensor error(X12)	Loader - 出料區-料匣到位檢(X12)-異常
10014	Loader - Outlet CV motor error(X13)	Loader - 出料區-料匣皮帶馬達(X13)-異常
10015	Loader - Boat fall down(X14)	Loader - Boat 盤入料檢(X14)-異常
10016	Loader - Boat fall down(X18)	Loader - Boat 盤出料檢(X18)-異常
10017	Loader - Track side CV motor error(X19)	Loader - Boat 盤皮帶馬達(X19)-異常
10018	Loader - Inlet area need Magazine(X21)	Loader - 入料區-料匣缺料檢知(X21)-異常
10019	Loader - Outlet area Full(X22)	Loader - 出料區-料匣滿料檢知(X22)-異常
10020	Loader - X position wrong can not move Z(X23)	Loader - Boat 盤 X 軸安全位置檢(X23)-異常
10021	Loader - Obstacles above the track(X24)	Loader - Boat 盤晶片平整檢 1(X24)-異常
10022	Loader - Obstacles above the track(X25)	Loader - Boat 盤晶片平整檢 2(X25)-異常
10023	Loader - Magazine Barcode Error	Loader - 料匣-BarCode-異常
10024	Loader - Boat Barcode Error	Loader - Boat-BarCode-異常
10025	Loader - Sensor X15 is on	Loader - Boat 盤位置檢 1(X15)-異常
10026	Loader - Sensor X16 is on	Loader - Boat 盤位置檢 2(X16)-異常
10027	Loader - Sensor X17 is on	Loader - Boat 盤位置檢 3(X17)-異常
10028	Loader - Boat Enter Error	Loader - Boat 入料異常
10029	Loader - Boat error not manually removed	Loader - Boat 盤異常未手動移出。

Equpiment Alarm - Sorter

ID	Name	Definition
10201	Sorter - Rail door inspection abnormal	Sorter - 門檢異常
10202	Sorter - The emergency stop was pressed.	Sorter - 緊急停止被按下
10203	Sorter - Air Pressure Low(X02)	Sorter - 氣壓不足(X02)
10204	Sorter - Vacuum Low(X03)	Sorter - 真空不足(X03)
10205	Sorter - Turrets does not at upper(X04)	Sorter - 吸筆不在上位(X04)
10206	Sorter - Unit missing(X05)	Sorter - 材料檢異常(X05)
10207	Sorter - Vacuum Low(X06)	Sorter - 真空檢異常(X06)
10208	Sorter - Door stop cylinder abnormal (X07)	Sorter - ST 區 Boat 盤門擋汽缸動檢(X07)-異常
10209	Sorter - Door stop cylinder abnormal (X08)	Sorter - ST 區 Boat 盤門擋汽缸原點檢(X08)-異常
10210	Sorter - Aside cylinder abnormal (X09)	Sorter - ST 區 Boat 盤靠邊汽缸動檢(X09)-異常
10211	Sorter - Aside cylinder abnormal (X10)	Sorter - ST 區 Boat 盤靠邊汽缸原點檢(X10)-異常
10212	Sorter - Door stop cylinder abnormal (X11)	Sorter - OK 區 Boat 盤門擋汽缸動檢(X11)-異常
10213	Sorter - Door stop cylinder abnormal (X12)	Sorter - OK 區 Boat 盤門擋汽缸原點檢(X12)-異常
10214	Sorter - Aside cylinder abnormal (X13)	Sorter - OK 區 Boat 盤靠邊汽缸動檢(X13)-異常
10215	Sorter - Aside cylinder abnormal (X14)	Sorter - OK 區 Boat 盤靠邊汽缸原點檢(X14)-異常
10216	Sorter - Door stop cylinder abnormal (X15)	Sorter - NG 區 Boat 盤門擋汽缸動檢(X15)-異常
10217	Sorter - Door stop cylinder abnormal (X16)	Sorter - NG 區 Boat 盤門擋汽缸原點檢(X16)-異常
10218	Sorter - Aside cylinder abnormal (X17)	Sorter - NG 區 Boat 盤靠邊汽缸動檢(X17)-異常
10219	Sorter - Aside cylinder abnormal (X18)	Sorter - NG 區 Boat 盤靠邊汽缸原點檢(X18)-異常
10220	Sorter - Boat fall down(X19)	Sorter - ST 區 Boat 盤入料檢(X19)-異常
10221	Sorter - Obstacles above the track(X21)	Sorter - ST 區 Boat 盤晶片平整檢 1(X21)-異常
10222	Sorter - Obstacles above the track(X24)	Sorter - ST 區 Boat 盤晶片平整檢 2(X24)-異常
10223	Sorter - Boat fall down(X26)	Sorter - ST 區 Boat 盤出料檢(X26)-異常
10224	Sorter - Boat fall down(X27)	Sorter - OK 區 Boat 盤入料檢(X27)-異常
10225	Sorter - Obstacles above the track(X29)	Sorter - OK 區 Boat 盤晶片平整檢 1(X29)-異常
10226	Sorter - Obstacles above the track(X32)	Sorter - OK 區 Boat 盤晶片平整檢 2(X32)-異常
10227	Sorter - Boat fall down(X34)	Sorter - OK 區 Boat 盤出料檢(X34)-異常
10228	Sorter - Boat fall down(X35)	Sorter - NG 區 Boat 盤入料檢(X35)-異常
10229	Sorter - Obstacles above the track(X37)	Sorter - NG 區 Boat 盤晶片平整檢 1(X37)-異常
10230	Sorter - Obstacles above the track(X40)	Sorter - NG 區 Boat 盤晶片平整檢 2(X40)-異常
10231	Sorter - Boat fall down(X42)	Sorter - NG 區 Boat 盤出料檢(X42)-異常

10232	Sorter - Outlet CV motor error(X43)	Sorter - ST 區-Boat 盤皮帶馬達(X43)-異常
10233	Sorter - Outlet CV motor error(X44)	Sorter - OK 區-Boat 盤皮帶馬達(X44)-異常
10234	Sorter - Outlet CV motor error(X45)	Sorter - NG 區-Boat 盤皮帶馬達(X45)-異常
10235	Sorter - Illegal IP	Sorter - 非法 IP
10236	Sorter - IP conflict	Sorter - IP 衝突
10237	Sorter - Port error	Sorter - Port 錯誤
10238	Sorter - Servere establishment failed	Sorter - Servere 建立失敗
10239	Sorter - Client connection failed	Sorter - Client 連線失敗
10240	Sorter - Pen Side Moter X Error	Sorter - 吸筆 X 軸馬達異常
10241	Sorter - Pen Side Moter Y Error	Sorter - 吸筆 Y 軸馬達異常
10242	Sorter - Pen Side Moter Z Error	Sorter - 吸筆 Z 軸馬達異常
10243	Sorter - Under side Moter T Error	Sorter - 底盤 Y 向馬達異常
10244	Sorter - Inbound and outbound detection detected objects	Sorter - 入出站檢偵測到物體
10245	Sorter - Abnormal height of pen	Sorter - 吸筆高度異常
10246	Sorter - Servo motor not Servo On	Sorter - CV 馬達異常
10247	Sorter - Servere establishment failed	Sorter - 伺服馬達未 Servo On
10248	Sorter - Call Unloader go to swap area fail	Sorter - 呼叫 Unloader 去交換區 失敗
10249	Sorter - Execute UnloadBoat fail	Sorter - 執行 UnLoadBoat 失敗
10250	Sorter - Find warping in OK area	Sorter - OK 區發現翹料
10251	Sorter - Find warping in NG area	Sorter - NG 區發現翹料
10252	Sorter - Find warping in ST area	Sorter - ST 區發現翹料

Equpiment Alarm - Unloader

ID	Name	Definition
10401	Unloader - Rail door inspection abnormal	Unloader - 門檢異常
10402	Unloader - The emergency stop was pressed.	Unloader - 緊急停止被按下
10403	Unloader - Air Pressure Low(X02)	Unloader - 氣壓不足(X02)
10404	Unloader - Jaws cylinder abnormal (X03)	Unloader - 夾爪氣缸動檢(X03)-異常
10405	Unloader - Jaws cylinder abnormal (X04)	Unloader - 夾爪氣缸原點檢(X04)-異常
10406	Unloader - Jaws cylinder X side abnormal (X05)	Unloader - 夾爪 X 軸氣缸動檢(X05)-異常
10407	Unloader - Jaws cylinder X side abnormal (X06)	Unloader - 夾爪 X 軸氣缸原點檢(X06)-異常
10408	Unloader - Jaws cylinder Z side abnormal (X07)	Unloader - 夾爪 Z 軸氣缸動檢(X07)-異常
10409	Unloader - Jaws cylinder Z side abnormal (X08)	Unloader - 夾爪 Z 軸氣缸原點檢(X08)-異常
10410	Unloader - Move stage cylinder abnormal (X09)	Unloader - 移載汽缸動檢(X09)-異常
10411	Unloader - Move stage cylinder abnormal (X10)	Unloader - 移載汽缸原點檢(X10)-異常
10412	Unloader - Boat fall down(X11)	Unloader - Boat 盤入料檢(X11)-異常
10413	Unloader - Boat fall down(X15)	Unloader - Boat 盤出料檢(X15)-異常
10414	Unloader - Track side CV motor error(X16)	Unloader - Boat 盤皮帶馬達(X16)-異常
10415	Unloader - Missing Magazine 1	Unloader - 料匣 1 到位檢(X17)-異常
10416	Unloader - Missing Magazine 2	Unloader - 料匣 2 到位檢(X18)-異常
10417	Unloader - Missing Magazine 3	Unloader - 料匣 3 到位檢(X19)-異常
10418	Unloader - Missing Magazine 4	Unloader - 料匣 4 到位檢(X20)-異常
10419	Unloader - Obstacles above the track(X22)	Unloader - Boat 盤晶片平整檢 1(X22)-異常
10420	Unloader - Obstacles above the track(X23)	Unloader - Boat 盤晶片平整檢 2(X23)-異常
10421	Unloader - Magazine 1 Missing Boat	Unloader - 料匣 1-Boat 盤在籍檢(X27)-異常
10422	Unloader - Magazine 2 Missing Boat	Unloader - 料匣 2-Boat 盤在籍檢((X28)-異常
10423	Unloader - Magazine 3 Missing Boat	Unloader - 料匣 3-Boat 盤在籍檢((X29)-異常
10424	Unloader - Magazine 4 Missing Boat	Unloader - 料匣 4-Boat 盤在籍檢((X30)-異常
10425	Unloader - Magazine Barcode Error	Unloader - 料匣-BarCode-異常
10426	Unloader - Boat Barcode Error	Unloader - Boat-BarCode-異常
10427	Unloader - Sensor X12 is on	Unloader - Boat 盤位置檢 1(X12)-異常
10428	Unloader - Sensor X13 is on	Unloader - Boat 盤位置檢 2(X13)-異常
10429	Unloader - Sensor X14 is on	Unloader - Boat 盤位置檢 3(X14)-異常
10430	Unloader - Boat Enter Error	Unloader - Boat 入料異常
10431	Unloader - Illegal IP	Unloader - 非法 IP

10432	Unloader - IP conflict	Unloader - IP 衝突
10433	Unloader - Port error	Unloader - Port 錯誤
10434	Unloader - Servere establishment failed	Unloader - Servere 建立失敗
10435	Unloader - Client connection failed	Unloader - Client 連線失敗
10436	Unloader - Pen Side Moter X Error	Unloader - Y 軸馬達異常
10437	Unloader - Pen Side Moter Z Error	Unloader - 乘載板 Z 軸馬達異常
10438	Unloader - The in-out material inspection detects an object or the cylinder is not in place	Unloader - 入出料檢偵測到物體或汽缸未到位
10439	Unloader - Unable to set motor parameters	Unloader - 無法設定馬達參數
10440	Unloader - The number of MAG layers set abnormally	Unloader - MAG 層數設定異常
10441	Unloader - Find warping	Unloader - 發現翹料

Equpiment Alarm - AOI

ID	Name	Definition
10601	AOI - Rail door inspection abnormal	AOI - 門檢異常
10602	AOI - The emergency stop was pressed	AOI - 緊急停止被按下
10603	AOI - Air Pressure Low(X02)	AOI - 氣壓不足(X02)
10604	AOI - Clip cylinder abnormal (X05)	AOI - 夾板氣缸(A)動檢(X05)-異常
10605	AOI - Clip cylinder abnormal (X06)	AOI - 夾板氣缸(A)原點檢(X06)-異常
10606	AOI - Clip cylinder abnormal (X07)	AOI - 夾板氣缸(B)動檢(X07)-異常
10607	AOI - Clip cylinder abnormal (X08)	AOI - 夾板氣缸(B)原點檢(X08)-異常
10608	AOI - Aside cylinder abnormal (X13)	AOI - 靠邊氣缸動檢(X13)-異常
10609	AOI - Aside cylinder abnormal (X14)	AOI - 靠邊氣缸原點檢(X14)-異常
10610	AOI - Boat fall down(X15)	AOI - Boat 盤入口下垂檢(X15)-異常
10611	AOI - Boat fall down(X16)	AOI - Boat 盤出口下垂檢(X16)-異常
10612	AOI - Obstacles above the track(X17)	AOI - Boat 盤晶片平整檢 1(X17)-異常
10613	AOI - Obstacles above the track(X18)	AOI - Boat 盤晶片平整檢 2(X18)-異常
10614	AOI - Track side CV motor error(X22)	AOI - Boat 盤皮帶馬達(X22)-異常
10615	AOI - Door stop cylinder abnormal (X23)	AOI - 門檔氣缸動檢-異常(X23)
10616	AOI - Door stop cylinder abnormal (X24)	AOI - 門檔氣缸原點檢-異常(x24)
10617	AOI - NG counter arrive setting	AOI - NG 數量達設定值
10701	AOI - Clamp A,B Initial retrack TimeOut	AOI - Clmap A 或 Clamp B 回原點逾時
10702	AOI - Z Axis Move TimeOut	AOI - Z 軸歸零時移動逾時
10703	AOI - Z Axis Homing TimeOut	AOI - Z 軸歸零逾時
10704	AOI - Initial Stopper Rising TimeOut	AOI - 初始化 Stopper 頂昇逾時
10705	AOI - X Axis Move 10mm TimeOut	AOI - X 軸移動 10mm 逾時
10706	AOI - Y Axis Move 10mm TimeOut	AOI - Y 軸移動 10mm 逾時
10707	AOI - X Axis Homing TimeOut	AOI - X 軸歸零逾時
10708	AOI - Y Axis Homing TimeOut	AOI - Y 軸歸零逾時
10801	AOI - Stopper Rising TimeOut	AOI - Stopper 升起逾時
10802	AOI - Load Product TimeOut	AOI - 產品在流道上入料逾時
10803	AOI - Load Product Arrial Stopper TimeOut	AOI - 產品到位逾時
10804	AOI - SidePusth stretch out TimeOut	AOI - 側推伸出失敗
10805	AOI - Clamp A or Clamp B stretch out TimeOut	AOI - 入料時 Clmap A 或 Clamp B 頂升逾時
10806	AOI - SidePush retract out TimeOut	AOI - 側推縮回逾時

10807	AOI - Clamp A or Clamp B retrack out TimeOut	AOI - 入料時 Clmap A 或 Clamp B 下降逾時
10808	AOI - Stopper retrack TimeOut	AOI - Stopper 縮回逾時
10809	AOI - Unload Product TimeOut	AOI - 出料逾時(產品未離開流道 Sensor)
10810	AOI - Stopper Rising TimeOut After Unload Product	AOI - 產品出料後 Stoppe 頂昇逾時
10901	AOI - AOI Move to Start Pos. of the row TimeOut	AOI - AOI 移動至單列起跑點逾時
10902	AOI - AOI Move to Row End Pos. of the row TimeOut	AOI - AOI 移動至單列結束點逾時

Event Report Link

Equipment event (p.28) will be transmitted with RTPID and SV, DV, EC values will be transmitted together, as shown below.

EventID	RPTID	SVID, DVID, ECID
MagazineInEvent(101)	101	MagazineID(4002)
BoatMoveInEvent(102)	102	BoatBarcode(8001)
PassBoatMoveOutEvent(103)	103	PassBoatBarcode(8002)
NGBoatMoveOutEvent(104)	104	NGBoatBarcode(8003)
EmptyBoatReleaseEvent(105)	105	EmptyBoatBarcode(8004)
Pass1MagazineMoveInEvent(106)	106	Pass1MagazineID(4003)
Pass2MagazineMoveInEvent(107)	107	Pass2MagazineID(4004)
NGMagazineMoveInEvent(108)	108	NGMagazineID(4005)
EmptyMagazineMoveInEvent(109)	109	EmptyMagazineID(4006)
Pass1MagazineMoveOutEvent(110)	110	LotID(4001) Pass1MagazineID(4003) Pass1MagazineUnitCount(8007)
Pass2MagazineMoveOutEvent(111)	111	LotID(4001) Pass2MagazineID(4004) Pass2MagazineUnitCount(8008)
NGMagazineMoveOutEvent(112)	112	LotID(4001) NGMagazineID(4005) NGMagazineUnitCount(8009)

BoatAssingNGBoatEvent(113)	113	LotID(4001) BoatAssingNGBoatBarcode(8005)
RemoteActionResultEvent(114)	114	RemoteActionResult(1011)
EquipmentStartRunEvent(115)	115	RecipeName(8006)

SECS-II Message List (H -> E)

Host → Equipment (Primary)		Equipment → Host (Secondary)	
Sn Fn		Sn F0	Abort Transaction
S1 F1	Are You There Request	S1 F2	On Line Data
S1 F3	Selected Equipment Status Request	S1 F4	Selected Equipment Status Data
S1 F11	Status Variable Name List Request	S1 F12	Status Variable name List Reply
S1 F13	Establish Communications Request	S1 F14	Establish Communications Request Acknowledge
S1 F15	Request Off-Line	S1 F16	Off-Line Acknowledge
S1 F17	Request On-Line	S1 F18	On-Line Acknowledge
S2 F13	Equipment Constant Request	S2 F14	Equipment Constant Data
S2 F15	New Equipment Constant Send	S2 F16	New Equipment Constant Acknowledge
S2 F17	Date and Time Request	S2 F18	Date and Time Data
S2 F23	Trace Initialize Send	S2 F24	Trace Initialize Acknowledge
S2 F25	Loop Back Diagnosis Request	S2 F26	Loop Back Diagnosis Reply
S2 F29	Equipment Constant Name List Request	S2 F30	Equipment Constant Name List Reply
S2 F31	Date and Time Send	S2 F32	Date and Time Acknowledge
S2 F33	Define Report	S2 F34	Define-Report Acknowledge
S2 F35	Link Event Report	S2 F36	Link Event Report Acknowledge
S2 F37	Enable/Disable Event Report	S2 F38	Enable/Disable Event Report Acknowledge
S2 F41	Remote Command Send	S2 F42	Remote Command Acknowledge
S5 F3	Enable/Disable Alarm Send	S5 F4	Enable/Disable Alarm Acknowledge
S5 F5	List Alarms Request	S5 F6	List Alarm Data
S5 F7	List Enabled Alarms Request	S5 F8	List Enabled Alarm Data
S6 F15	Event Report Request	S6 F16	Event Report Data
S6 F19	Individual Report Request	S6 F20	Individual Report Data
S7F1	Process Program Load Inquire	S7F2	Process Program Load Grant
S7F5	Process Program Request	S7F6	Process Program Data
S7F19	Current Process Program Dir. Request	S7F20	List Process Program Data
S7F25	Formatted Process Program Request	S7F26	Formatted Process Program Data
S10 F3	Terminal Display	S10 F4	Terminal Display Acknowledge

SECS-II Message List (E -> H)

Host → Equipment (Secondary)		Equipment → Host (Primary)	
Sn Fn		Sn F0	Abort Transaction
S1 F2	On Line Data	S1 F1	Are You There Request
S1 F14	Establish Communications Request Acknowledge	S1 F13	Establish Communications Request
S2 F18	Date and Time Data	S2 F17	Date and Time Request
S5 F2	Alarm Report Acknowledge	S5 F1	Alarm Report Send
S6 F2	Trace Data Acknowledge	S6 F1	Trace Data Send
S6F12	Event Report Acknowledge	S6F11	Event Report Send
		S9 F7	Illegal Data
		S9 F9	Transaction Timer Timeout
S10 F2	Terminal Request Acknowledge	S10 F1	Terminal Request

SECS-II Message Example (Host perspective)

S1F1 : Are You There (H -> E)

Description **:**

The host may send this message to the EQ at any time. EQ sends this message periodically as a “heartbeat” to determine if the SECS link is alive.

```
2022/2/24 10:06:33.632: Send S1F1 <Header: 00 00 81 01 00 00 00 02 B0 E2>
<L[1]
  <A[14] "Are you there?">
>
.
```

S1F2 : On-Line Data (E -> H)

Description **:**

EQ reports its Model Number and Software Revision to Host

```
2022/2/24 10:06:33.633: Receive S1F2 <Header: 00 00 01 02 00 00 00 02 B0 E2>
<L[2]
  <A[13] "SMT Equipment">
  <A[7] "1.0.0.0">
>
.
```

.

S1F3 : Selected Status Request (H -> E)

Description :

A request to EQ to report selected values of its status.

```
2022/2/24 10:08:19.665: Send S1F3 <Header: 00 00 81 03 00 00 00 02 B0 ED>
<U4[3] 100001 2002 2003>
.
```

S1F4 : Selected Status Data (E -> H)

Description :

The equipment reports the value of each SVID requested in the order requested.

```
2022/2/24 10:08:19.665: Receive S1F4 <Header: 00 00 01 04 00 00 00 02 B0 ED>
<L[3]
 <U1[1] 0>
 <A[0] "">
 <F4[1] 0>
>
.
```

S1F11 : Status Variable NamelistRequest (H -> E)

Description :

A request to the equipment to identify certain status variables.

```
2022/2/24 10:10:52.536: Send S1F11 <Header: 00 00 81 0B 00 00 00 02 B0 EF>
<L[2]
 <U4[1] 2002>
 <U4[1] 2003>
>
.
```

S1F12 : Status Variable Namelist Reply (E -> H)

Description :

EQ reports to the Host the name and units of the requested SVs.

```
2022/2/24 10:10:52.538: Receive S1F12 <Header: 00 00 01 0C 00 00 00 02 B0 EF>
<L[2]
 <L[3]
  <U4[1] 2002>
  <A[19] "SV_CONTROL_JOB_NAME">
  <A[0] "">
 >
 <L[3]
  <U4[1] 2003>
  <A[17] "SV_INMP1_NOW_TEMP">
  <A[6] "Degree">
 >
>
.
```

S1F13 : Establish Communication Request (H -> E)

Description:
Either end of the link may send S1F13 as the first message to establish connection of the link. Under EQ's initialization logic, EQ will always attempt to establish the link. If no Host acceptance reply is received. EQ sends this message periodically.

```
2022/2/24 10:12:02.624: Send S1F13 <Header: 00 00 81 0D 00 00 00 02 B0 F0>
<L[0]
>
.
```

S1F14 : Establish Communication Request Acknowledge (E -> H)

Description:
Accept or deny Establish Communication Request.

```
2022/2/24 10:12:02.625: Receive S1F14 <Header: 00 00 01 0E 00 00 00 02 B0 F0>
<L[2]
 <B[1] 0x00>
 <L[2]
  <A[13] "SMT Equipment">
  <A[7] "1.0.0.0">
 >
>
.
```

S1F15 : Request Off-Line (H -> E)

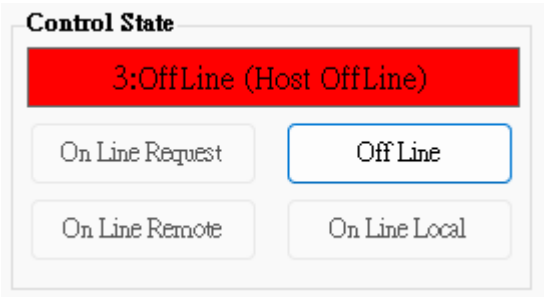
Description:
Host requests EQ to go Off-Line

```
2022/2/24 10:12:48.521: Send S1F15 <Header: 00 00 81 0F 00 00 00 02 B0 F1>
.
```

S1F16 : Off-Line Acknowledge (E -> H)

Description:
EQ control state transits to Host Off-Line and sends this message in response to S1F15 primary.

```
2022/2/24 10:12:48.522: Receive S1F16 <Header: 00 00 01 10 00 00 00 02 B0 F1>
<B[1] 0x00>
.
```



S1F17 : Request On-Line (H -> E)

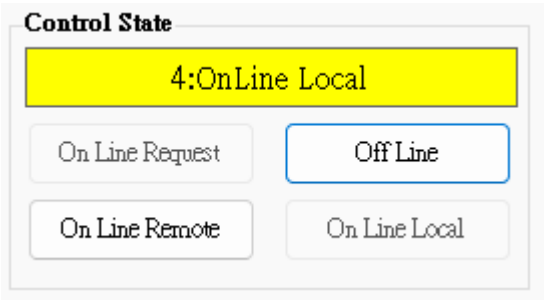
Description:
Host requests EQ to go On-Line

```
2022/2/24 10:14:33.960: Send S1F17 <Header: 00 00 81 11 00 00 00 02 B0 F2>
.
```

S1F18 : On-Line Acknowledge (E -> H)

Description:
EQ responds to S1F17 primary in an attempt to go On-Line. Data item ONLACK indicates the success of failure of the attempt.

```
2022/2/24 10:14:33.961: Receive S1F18 <Header: 00 00 01 12 00 00 00 02 B0 F2>
<B[1] 0x00>
.
```



S2F13 : Equipment Constant Request (H -> E)

Description:
Constants and other values that are changed infrequently can be obtained using this message.

```
2022/2/24 10:04:22.056: Send S2F13 <Header: 00 00 82 0D 00 00 00 02 B0 E0>
<L[3]
 <U4[1] 1001>
 <U4[1] 1002>
 <U4[1] 1003>
>
.
```

S2F14 : Equipment Constant Data (E -> H)

Description:
EQ returns constant data in the order requested in S2F13.

```
2022/2/24 10:04:22.057: Receive S2F14 <Header: 00 00 02 0E 00 00 00 02 B0 E0>
<L[3]
 <U1[1] 1>
 <U2[1] 50>
 <F4[1] 3.456>
>
.
```

S2F15 : New Equipment Constant Send (H -> E)

Description:
Change one or more equipment constants.

```
2022/2/24 10:02:43.540: Send S2F15 <Header: 00 00 82 0F 00 00 00 02 B0 DF>
<L[3]
 <L[2]
  <U4[1] 1002>
  <U2[1] 50>
 >
 <L[2]
  <U4[1] 1003>
  <F4[1] 3.456>
 >
 <L[2]
  <U4[1] 1011>
  <A[40] "_123456789_123456789_123456789_123456789">
 >
>
.
```

S2F16 : Equipment Constant Send Acknowledge (E -> H)

Description:
Change one or more equipment constants.

```
2022/2/24 10:02:43.581: Receive S2F16 <Header: 00 00 02 10 00 00 00 02 B0 DF>
<B[1] 0x00>
.
```

EAC	Description
0x00	OK.
0x01	At least one ECID is invalid.
0x03	At least one ECV is out of range.

S2F17 : Date and Time Request (H -> E)

Description:
Host sends this message at any time to determine the date and time base EQ is currently using.

```
2022/2/24 10:15:34.543: Send S2F17 <Header: 00 00 82 11 00 00 00 02 B0 F4>
.
```

S2F18 : Date and Time Data (E -> H)

Description:
Host sends this message at any time to determine the date and time base EQ is currently using.

```
2022/2/24 10:15:34.543: Send S2F17 <Header: 00 00 82 11 00 00 00 02 B0 F4>
.
2022/2/24 10:15:34.544: Receive S2F18 <Header: 00 00 02 12 00 00 00 02 B0 F4>
<A[16] "2022022410153400">
.
```

S2F23 : Trace Initialize Send (H -> E)

Description:
Trace initial send to device to enable/disable trace initial data. The trace data is returned on (S6F1) and is related to the original request by the TRID.

```
2022/2/24 10:35:34.704: Send S2F23 <Header: 00 00 82 17 00 00 00 02 B0 F5>
<L[5]
 <U4[1] 123456789>
 <A[6] "000003">
 <U4[1] 27>
 <U4[1] 3>
 <L[3]
  <U4[1] 100001>
  <U4[1] 2002>
  <U4[1] 2003>
 >
>
.
```

S2F24 : Trace Initialize Acknowledge (E -> H)

Description:
Acknowledge or error for Trace Initialize Send(S2F23)

```
2022/2/24 10:35:34.707: Receive S2F24 <Header: 00 00 02 18 00 00 00 02 B0 F5>
<B[1] 0x00>
.
```

TIAACK	Description
0x00	Normal. Everything correct.
0x03	Invalid period. DSPER is incorrect. The Equipment does not start the trace.

S2F25 : Diagnostic Loopback Request (H -> E)

Description:
Host sends an arbitrary binary string to test operation of the link.

```
2022/2/24 10:41:10.503: Send S2F25 <Header: 00 00 82 19 00 00 00 02 B0 F6>
<B[10] 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09>
.
```

S2F26 Diagnostic Loopback Data (E -> H)

Description:
EQ “echoes” back the binary data from Host’s S2F25.

```
2022/2/24 10:41:10.504: Receive S2F26 <Header: 00 00 02 1A 00 00 00 02 B0 F6>
<B[10] 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09>
.
```

S2F29 Equipment Constant Namelist Request (H -> E)

Description:
This function allows the host to retrieve basic information about what equipment constants are available in the equipment.

```
2022/2/24 10:46:20.590: Send S2F29 <Header: 00 00 82 1D 00 00 00 02 B0 F8>
<L[2]
 <U4[1] 1001>
 <U4[1] 1003>
>
.
```

S2F30 : Equipment Constant Namelist Reply (E -> H)

Description:
Data Response for Equipment Constant Namelist Request(S2F29).

```
2022/2/24 10:46:20.590: Receive S2F30 <Header: 00 00 02 1E 00 00 00 02 B0 F8>
<L[2]
 <L[6]
  <U4[1] 1001>
  <A[10] "EC1_UINT_1">
  <A[1] "0">
  <A[1] "1">
  <A[1] "1">
  <A[0] "">
 >
 <L[6]
  <U4[1] 1003>
  <A[8] "EC3_FT_4">
  <A[3] "0.1">
  <A[4] "10.9">
  <A[5] "3.456">
  <A[6] "Degree">
 >
>
.
```

S2F31 : Date and Time Send (H -> E)

Description:
Useful to synchronize the equipment time with the host time base.

```
2022/2/24 10:52:10.325: Send S2F31 <Header: 00 00 82 1F 00 00 00 02 B1 06>
<A[16] "2013081711223300">
.
```

S2F32 Date and Time Acknowledge (E -> H)

Description:
Acknowledge the receipt of time and date.

```
2022/2/24 10:52:10.326: Receive S2F32 <Header: 00 00 02 20 00 00 00 02 B1 06>
<B[1] 0x01>
.
```

TIACK	Description
0x00	Normal. Everything correct.
0x01	Invalid Date and/or Time.

S2F33 Define Report (H -> E)

Description:
The purpose of this message is for the host to define a group of reports for the equipment. The event report will be sent via S6F11 when the event occurred.

```
2022/2/24 10:55:47.918: Send S2F33 <Header: 00 00 82 21 00 00 00 02 B1 08>
<L[2]
 <U1[1] 0>
 <L[1]
  <L[2]
   <U4[1] 7001>
   <L[3]
    <U4[1] 17>
    <U4[1] 100001>
    <U4[1] 2002>
   >
  >
 >
>
.
```

S2F34 Define Report Acknowledge (E -> H)

Description:
Acknowledge or error if an error condition is detected the entire message is rejected(i.e., partial changes are not allowed)

```
2022/2/24 10:55:47.932: Receive S2F34 <Header: 00 00 02 22 00 00 00 02 B1 08>
<B[1] 0x00>
.
```

S2F35 : Link Event Report (H -> E)

Description:
The purpose of this message is for the host to link n reports to an event(CEID). These linked event reports will default to 'disabled' upon linking. That is, the occurrence of an event would not cause the report to be sent until enabled. See S2F37 for enabling reports.

```
2022/2/24 11:00:04.566: Send S2F35 <Header: 00 00 82 23 00 00 00 02 B1 09>
<L[2]
 <U1[1] 0>
 <L[1]
  <L[2]
   <U4[1] 101>
   <L[1]
    <U4[1] 7001>
   >
  >
 >
>
.
```

S2F36 Link Event Report Acknowledge (E -> H)

Description:
Acknowledge or error if an error condition is detected the entire message is rejected(i.e., partial changes are not allowed).

```
2022/2/24 11:00:04.576: Receive S2F36 <Header: 00 00 02 24 00 00 00 02 B1 09>
<B[1] 0x00>
.
```

S2F37 : Enable/Disable Event Report (H -> E)

Description:
The purpose of this message is for the host to enable or disable reporting for a group of events(CEIDs).

```
2022/2/24 11:02:00.886: Send S2F37 <Header: 00 00 82 25 00 00 00 02 B1 0A>
<L[2]
 <Boolean[1] 0x1>
 <L[2]
  <U4[1] 101>
  <U4[1] 102>
 >
>
.
```

S2F38 : Enable/Disable Event Report Acknowledge (E -> H)

Description:
Acknowledge or error if an error condition is detected the entire message is rejected(i.e., partial changes are not allowed).

```
2022/2/24 11:02:00.887: Receive S2F38 <Header: 00 00 02 26 00 00 00 02 B1 0A>
<B[1] 0x00>
.
```

ERACK	Description
0x00	OK.
0x01	Denied. At least one CEID does not exist.

S2F41 : Remote Command Send (H -> E)

Description:
The host requests the equipment perform the specified remote command with the associated parameters.

```
2022/2/24 13:54:12.874: Send S2F41 <Header: 00 00 82 29 00 00 00 6D 0A 29>
<L[2]
 <A[9] "PP-SELECT">
 <L[2]
  <L[2]
   <A[4] "PPID">
   <A[3] "PP1">
  >
  <L[2]
   <A[6] "PortID">
   <U1[1] 1>
  >
 >
>
.
```

RCMD String	Function
"START"	Starts a Process Program.
"ABORT"	Aborts a running Process Program.
"PPSELECT"	Pauses a running Process Program.
" LOCAL "	On-Line/LOCAL
" REMOTE "	On-Line/REMOTE

S2F42 : Remote Command Acknowledge (E -> H)

Description:

Acknowledge host's command or error. If command is not accepted due to one or more invalid parameters, then a list of invalid parameters will be returned containing the parameter name and reason for being invalid.

```
2022/2/24 13:54:12.875: Receive S2F42 <Header: 00 00 02 2A 00 00 00 6D 0A 29>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

HCACK	Description
0x00	OK. All normal.
0x01	Invalid command. Equipment rejects command.
0x02	Cannot perform now. Equipment rejects command.
0x03	At least one parameter is invalid. Equipment rejects the command.
0x40	Control State is Local. Equipment rejects command.
0x41	Incorrect Process State for this command. Equipment rejects command.
0x42	Process Program contains an invalid format. Unable to load program.

S5F1 : Alarm Report (E -> H)

Description:
This message reports a change in or presence of an alarm condition. One message will be issued when the alam is set and one message will be issued when the alarm is cleared. Irrecoverable errors and attention flags may not have a corresponding clear message.

```
2022/2/24 15:58:45.159: Receive S5F1 <Header: 00 00 85 01 00 00 00 97 D9 75>
<L[3]
 <B[1] 0x80>
 <U4[1] 9001>
 <A[17] "Pump Press Fressl">
>
.
```

S5F2 : Alarm Acknowledge (H -> E)

Description:
The host acknowledges the alarm report. This equipment ignores the ACKC5.

```
2022/2/24 15:58:45.162: Send S5F2 <Header: 00 00 05 02 00 00 00 97 D9 75>
<B[1] 0x00>
.
```

S5F3 Enable/Disable Alarm Send (H -> E)

Description:
This message will change the state of the enable bit in the equipment. The enable bit determines if the alarm will be sent to the host. Alarms which are not controllable in this way are unaffected by this message.

```
2022/2/24 14:19:08.561: Send S5F3 <Header: 00 00 85 03 00 00 00 48 85 D9>
<L[2]
 <B[1] 0x80>
 <U4[1] 9001>
>
.
```

ALED	Description
0x80	Enable sending alarm.
0x00	Disable sending alarm.

S5F4 Enable/Disable Alarm Acknowledge (E -> H)

Description:
The host acknowledges the alarm report. This equipment ignores the ACKC5.

```
2022/2/24 14:19:08.565: Receive S5F4 <Header: 00 00 05 04 00 00 00 48 85 D9>
<B[1] 0x00>
.
```

ACKC5	Description
0x00	Normal. Everything correct.
0x01	Invalid ALID.

S5F5 : List Alarms Request (H -> E)

Description:
This message requests the equipment to send binary and analog alarm information to the host.

```
2022/2/24 14:23:57.458: Send S5F5 <Header: 00 00 85 05 00 00 00 48 85 DD>
<U4[3] 9001 9002 9003>
.
```

S5F6 : List Alarm Data (E -> H)

Description:
This message contains the alarm data known to the equipment.

```
2022/2/24 14:36:36.060: Receive S5F6 <Header: 00 00 05 06 00 00 00 2B 11 8B>
<L[3]
 <L[3]
  <B[1] 0x00>
  <U4[1] 9001>
  <A[17] "Pump Press Fressl">
 >
 <L[3]
  <B[1] 0x00>
  <U4[1] 9002>
  <A[22] "Emergency Stop pressed">
 >
 <L[3]
  <B[1] 0x00>
  <U4[1] 9003>
  <A[16] "Heater High Fail">
 >
>
.
```

S5F7 : List Enabled Alarms Request (H -> E)

Description:
List all alarms which are enabled .

```
2022/2/24 14:36:51.378: Send S5F7 <Header: 00 00 85 07 00 00 00 2B 11 8C>
.
```

S5F8 : List Enabled Alarm Data (E -> H)

Description:
This message is similar to S5F6 except that it lists only alarms which are enabled.

```
2022/2/24 14:36:51.384: Receive S5F8 <Header: 00 00 05 08 00 00 00 2B 11 8C>
<L[4]
 <L[3]
  <B[1] 0x00>
  <U4[1] 9001>
  <A[17] "Pump Press Fressl">
 >
 <L[3]
  <B[1] 0x00>
  <U4[1] 9002>
  <A[22] "Emergency Stop pressed">
 >
 <L[3]
  <B[1] 0x00>
  <U4[1] 9003>
  <A[16] "Heater High Fail">
 >
 <L[3]
  <B[1] 0x00>
  <U4[1] 1000001>
  <A[16] "Heater Low Fail ">
 >
>
.
```

S6F1 : Trace Data Send (E -> H)

Description:
This function sends samples to the host according to the trace setup done by S2F23.
Trace is a time-driven form of equipment status.

```
2022/2/24 16:10:43.192: Receive S6F1 <Header: 00 00 06 01 00 00 00 53 F2 B2>
<L[4]
 <U4[1] 123456789>
 <U4[1] 3>
 <A[16] "2022022416104300">
 <L[9]
  <U1[1] 0>
  <A[0] "">
  <F4[1] 0>
  <U1[1] 0>
  <A[0] "">
  <F4[1] 0>
  <U1[1] 0>
  <A[0] "">
  <F4[1] 0>
 >
>
.
```

S6F2 : Trace Data Acknowledge (H -> E)

Description:
List all alarms which are enabled .

```
2022/2/24 16:10:43.200: Send S6F2 <Header: 00 00 06 02 00 00 00 53 F2 B2>
.
```

S6F11 : Event Report Send (E -> H)

Description:
The purpose of this message is for the equipment to send a defined, linked, and enabled group of reports to the host upon the occurrence of an event(CEID).

```
2022/2/24 16:12:46.967: Receive S6F11 <Header: 00 00 86 0B 00 00 00 53 F2 BB>
<L[3]
 <U2[1] 1>
 <U4[1] 101>
 <L[1]
  <L[2]
   <U4[1] 7001>
   <L[2]
    <U1[1] 0>
    <A[0] "">
   >
  >
 >
>
.
```

S6F12 : Event Report Acknowledge (H -> E)

Description:
Acknowledge or error for Event Report Send(S6F11).

```
2022/2/24 16:12:46.971: Send S6F12 <Header: 00 00 06 0C 00 00 00 53 F2 BB>
<B[1] 0x00>
.
```

S6F15 : Event Report Request (H -> E)

Description:
The purpose of this message is for the host to demand a given report group form the equipment.

```
2022/2/24 14:41:39.817: Send S6F15 <Header: 00 00 86 0F 00 00 00 2B 11 91>
<U4[1] 103>
.
```

S6F16 : Event Report Data (E -> H)

Description:
EQ sends reports linked to given CEID to host.

```
2022/2/24 14:41:39.817: Receive S6F16 <Header: 00 00 06 10 00 00 00 2B 11 91>
<L[3]
 <U2[1] 0>
 <U4[1] 103>
 <L[1]
  <L[2]
   <U4[1] 7004>
   <L[11]
    <F4[1] 0>
    <U1[1] 0>
    <U2[1] 0>
    <U4[1] 0>
    <F4[1] 0>
    <I1[1] 0>
    <I2[1] 0>
    <I4[1] 0>
    <A[0] "">
    <Boolean[1] 0x0>
    <U1[1] 0>
   >
  >
 >
>
.
```

S6F19 : Individual Report Request (H -> E)

Description:
Host requests a Report for the specified RPTID.

```
2022/2/24 14:44:55.416: Send S6F19 <Header: 00 00 86 13 00 00 00 2B 11 93>
<U4[1] 7004>
.
```

S6F20 : Individual Report Data Event Report Data (E -> H)

Description:
EQ sends variable data defined for the given to the host.

```
2022/2/24 14:44:55.416: Receive S6F20 <Header: 00 00 06 14 00 00 00 2B 11 93>
<L[11]
 <F4[1] 0>
 <U1[1] 0>
 <U2[1] 0>
 <U4[1] 0>
 <F4[1] 0>
 <I1[1] 0>
 <I2[1] 0>
 <I4[1] 0>
 <A[0] "">
 <Boolean[1] 0x0>
 <U1[1] 0>
>
.
```

S7F1 : Process Program Load Inquire (H -> E)

Description:
The sender requests permission to send a process program.

```
2022/2/24 14:51:29.744: Send S7F1 <Header: 00 00 87 01 00 00 00 2B 11 96>
<L[2]
 <A[3] "PP1">
 <U4[1] 21>
>
.
```

S7F2 : Process Program Load Grant (E -> H)

Description:
The receiver grants permission to send a process program.

```
2022/2/24 14:51:29.744: Receive S7F2 <Header: 00 00 07 02 00 00 00 2B 11 96>
<B[1] 0x00>
.
```

PPGNT	Description
0x00	Normal. Permission granted.
0x01	Already have
0x02	No space
0x03	Invalid PPID
0x04	Busy.
0x05	Will not accept
0x06	Other error

S7F5 : Process Program Request (H -> E)

Description:
Request the other end of the link to send a specified process program from its library.

```
2022/2/24 14:53:38.416: Send S7F5 <Header: 00 00 87 05 00 00 00 2B 11 99>
<A[3] "PP1">
.
```

S7F6 : Process Program Data (E -> H)

Description:
Send the requested process program.

```
2022/2/24 14:53:38.417: Receive S7F6 <Header: 00 00 07 06 00 00 00 2B 11 99>
<L[2]
 <A[3] "PP1">
 <A[21] "This is a test recipe">
>
.
```

S7F19 : Process Program Dir. Request (H -> E)

Description:
The host requests a directory of the process program currently in the EQ's library.

```
2022/2/24 14:55:09.647: Send S7F19 <Header: 00 00 87 13 00 00 00 2B 11 9A>
.
```

S7F20 : Current Process Program Data (E -> H)

Description:
The EQ sends a directory of its process program library. If the library is empty, the list will be zero length.

```
2022/2/24 14:55:09.649: Receive S7F20 <Header: 00 00 07 14 00 00 00 2B 11 9A>
<L[12]
 <A[4] "1234">
 <A[2] "20">
 <A[3] "222">
 <A[2] "99">
 <A[3] "PP1">
 <A[13] "PP1_Formatted">
 <A[3] "PP2">
 <A[7] "PP2.pjb">
 <A[13] "PP2_Formatted">
 <A[47] "PP3_1234567890_1234567890_1234567890_1234567890">
 <A[10] "PP3_BINARY">
 <A[13] "PP4_Formatted">
>
.
```

S7F25 : Formatted Process Program Request (H -> E)

Description:

The host requests a process program currently in the EQ's library.

```
2022/2/24 14:56:37.776: Send S7F25 <Header: 00 00 87 19 00 00 00 2B 11 A1>
<A[13] "PP1_Formatted">
.
```

S7F26 : Formatted Process Program Data (E -> H)

Description:

The EQ sends a formatted process program data to the host.

```
2022/2/24 14:56:37.778: Receive S7F26 <Header: 00 00 07 1A 00 00 00 2B 11 A1>
<L[4]
 <A[13] "PP1_Formatted">
 <A[13] "SMT Equipment">
 <A[7] "1.0.0.0">
 <L[1]
  <L[2]
   <U2[1] 2>
   <L[9]
    <U1[1] 0>
    <U2[1] 1>
    <U4[1] 1>
    <I1[1] 100>
    <I2[1] 1000>
    <I4[1] 1000000>
    <F4[1] 0.1>
    <Boolean[1] 0x0>
    <A[50] "A123456789_123456789_123456789_123456789_123456789">
   >
  >
 >
>
.
```

S9F7 : Invalid Data

Description:
This message indicates that the stream and function were recognized, but the associated data format could not be interpreted.

```
2022/2/24 16:20:14.221: Receive S9F7 <Header: 00 00 09 07 00 00 00 09 78 97>
<B[10] 0x00 0x00 0x86 0x01 0x00 0x00 0x00 0x7E 0xF9 0xEE>
.
```

S9F9 : Transaction Timer Timeout

Description:
This message indicates that a transaction (receive) timer has timed out and that the corresponding transaction has been aborted. It is up to the host to respond to this error in an appropriate manner to keep the system operational.

```
16:21:03.257: T3 Timeout(8321520).
2022/2/24 16:21:03.258: Send S9F9 <Header: 00 00 09 09 00 00 00 7E F9 F1>
<B[10] 0x00 0x00 0x09 0x09 0x00 0x00 0x00 0x7E 0xF9 0xEE>
.
```

S10F1 : Terminal Request

Description:
A terminal text message to the host.

```
2022/2/24 16:26:53.637: Receive S10F1 <Header: 00 00 8A 01 00 00 00 09 78 9F>
<L[2]
 <B[1] 0x00>
 <A[26] "Hi ! This is Eqp's Message">
>
.
```

S10F2 : Terminal Request Acknowledge

Description:
Acknowledge or error for terminal request (S10F1).

```
2022/2/24 16:26:53.639: Send S10F2 <Header: 00 00 0A 02 00 00 00 09 78 9F>
<B[1] 0x00>
.
```

S10F3 : Terminal Display, Single (H -> E)

Description:
Data to be displayed.

```
2022/2/24 15:33:09.688: Send S10F3 <Header: 00 00 8A 03 00 00 00 2B 11 A2>
<L[2]
 <B[1] 0x00>
 <A[16] "TerminalDataTest">
>
.
```

TID	Description
Any	Equipment displays the text on its CRT.

Message	Description
Any	Equipment displays the text
0XX	File MappingTable in folder ini or EQ Set

S10F4 : Terminal Request Acknowledge (E -> H)

Description:
Acknowledge or error for terminal request (S10F3).

```
2022/2/24 15:33:09.689: Receive S10F4 <Header: 00 00 0A 04 00 00 00 2B 11 A2>
<B[1] 0x00>
.
```

ACKC10	Description
0x00	Normal.

PPBODY – AOI

ID	Name	Enable	Unit	Format	Min	Max	Explanation
101	AOI_CornerA_X	TRUE	mm	ASCII	0	500	Boat A 角落的 X 位置
102	AOI_CornerA_Y	TRUE	mm	ASCII	0	500	Boat A 角落的 Y 位置
103	AOI_CornerC_X	TRUE	mm	ASCII	0	500	Boat C 角落的 X 位置
104	AOI_CornerC_Y	TRUE	mm	ASCII	0	500	Boat C 角落的 Y 位置
114	AOI_Substrate_Row	TRUE	Number	ASCII	1	2	產品參數
115	AOI_Substrate_Col	TRUE	Number	ASCII	1	1	產品參數
116	AOI_Block_Row	TRUE	Number	ASCII	1	4	產品參數
117	AOI_Block_Col	TRUE	Number	ASCII	1	4	產品參數
118	AOI_Unit_Row	TRUE	Number	ASCII	2	20	產品參數
119	AOI_Unit_Col	TRUE	Number	ASCII	2	20	產品參數
120	AOI_Boat_Left_Offset	TRUE	pixel	ASCII	0	3500	Boat 偏移量
121	AOI_Boat_Right_Offset	TRUE	pixel	ASCII	0	3500	Boat 偏移量
122	AOI_Boat_Top_Offset	TRUE	pixel	ASCII	0	2500	Boat 偏移量
123	AOI_Boat_Bottom_Offset	TRUE	pixel	ASCII	0	2500	Boat 偏移量

PPBODY - Sorter

ID	Name	Enable	Unit	Format	Min	Max	Explanation
201	Sorter_Recipe_Name	TRUE	Null	A			Recipe 名稱
202	Sorter_Recipe_ProductName	TRUE	Null	A			產品名稱
203	Sorter_Recipe_X_Num	TRUE	Number	A	1	20	橫向料數
204	Sorter_Recipe_Y_Num	TRUE	Number	A	1	10	縱向料數
205	Sorter_Recipe_Board1_P1_X	TRUE	mm	A	41000	42500	NG 區軌道 P1 點 X
206	Sorter_Recipe_Board1_P1_Y	TRUE	mm	A	551500	552500	NG 區軌道 P1 點 Y
207	Sorter_Recipe_Board1_P1_Z	TRUE	mm	A	73500	74500	NG 區軌道 P1 點 Z
208	Sorter_Recipe_Board1_P4_X	TRUE	mm	A	311000	312000	NG 區軌道 P4 點 X
209	Sorter_Recipe_Board1_P4_Y	TRUE	mm	A	672000	673000	NG 區軌道 P4 點 Y
210	Sorter_Recipe_Board1_P4_Z	TRUE	mm	A	73500	74500	NG 區軌道 P4 點 Z
211	Sorter_Recipe_Board2_P1_X	TRUE	mm	A	41000	42500	OK 區軌道 P1 點 X
212	Sorter_Recipe_Board2_P1_Y	TRUE	mm	A	325500	327500	OK 區軌道 P1 點 Y
213	Sorter_Recipe_Board2_P1_Z	TRUE	mm	A	73500	74500	OK 區軌道 P1 點 Z
214	Sorter_Recipe_Board2_P4_X	TRUE	mm	A	311000	312000	OK 區軌道 P4 點 X
215	Sorter_Recipe_Board2_P4_Y	TRUE	mm	A	445500	447500	OK 區軌道 P4 點 Y
216	Sorter_Recipe_Board2_P4_Z	TRUE	mm	A	73500	74500	OK 區軌道 P4 點 Z
217	Sorter_Recipe_Board3_P1_X	TRUE	mm	A	41000	42500	ST 區軌道 P1 點 X
218	Sorter_Recipe_Board3_P1_Y	TRUE	mm	A	99000	101000	ST 區軌道 P1 點 Y
219	Sorter_Recipe_Board3_P1_Z	TRUE	mm	A	73500	74500	ST 區軌道 P1 點 Z
220	Sorter_Recipe_Board3_P4_X	TRUE	mm	A	311000	313000	ST 區軌道 P4 點 X
221	Sorter_Recipe_Board3_P4_Y	TRUE	mm	A	220000	221000	ST 區軌道 P4 點 Y
222	Sorter_Recipe_Board3_P4_Z	TRUE	mm	A	73500	74500	ST 區軌道 P4 點 Z
223	Sorter_Recipe_Zdown_Position	TRUE	mm	A	80000	90000	下壓高度

MAGAZINE MOVEIN LOG

Description:

MagazineID attached to the MagazineMoveInEvent(101).

```
2022/3/7 11:07:52.482: Receive S6F11 <Header: 00 00 86 0B 00 00 00 2D E3 47>
<L[3]
 <U2[1] 1>
 <U4[1] 101>
 <L[1]
  <L[2]
   <U4[1] 101>
   <L[1]
    <A[9] "MA12965WE">
   >
  >
 >
>
.
2022/3/7 11:07:52.486: Send S6F12 <Header: 00 00 06 0C 00 00 00 2D E3 47>
<B[1] 0x00>
.
2022/3/7 11:08:16.266: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[15] "MAGAZINE_MOVEIN">
 <L[3]
  <L[2]
   <A[6] "LOT_ID">
   <A[10] "SD58767868">
  >
  <L[2]
   <A[11] "MAGAZINE_ID">
   <A[9] "MA12965WE">
  >
  <L[2]
   <A[22] "MAGAZINE_MOVEIN_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/3/7 11:08:16.271: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with MAGAZINE_MOVEIN_STATUS

BOAT MOVEIN LOG

Description:

BoatBarcode attached to the BoatMoveInEvent(102).

```
2022/5/17 09:54:01.779: Receive S6F11 <Header: 00 00 86 0B 00 00 00 6B DB F3>
<L[3]
 <U2[1] 7>
 <U4[1] 102>
 <L[1]
  <L[2]
   <U4[1] 102>
   <L[1]
    <A[23] "67.5x75-T-014-014-N0191">
   >
  >
 >
>
.
2022/5/17 09:54:01.783: Send S6F12 <Header: 00 00 06 0C 00 00 00 6B DB F3>
<B[1] 0x00>
.
```

```
2022/5/17 09:54:03.092: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[11] "BOAT_MOVEIN">
 <L[2]
  <L[2]
   <A[12] "BOAT_BARCODE">
   <A[23] "67.5x75-T-014-014-N0191">
  >
  <L[2]
   <A[18] "BOAT_MOVEIN_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/5/17 09:54:03.092: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with BOAT_MOVEIN_STATUS

PASSBOAT MOVEOUT LOG

Description:

PassBoatBarcode attached to the PassBoatMoveOutEvent(103).

```
2022/3/7 11:26:32.851: Receive S6F11 <Header: 00 00 86 0B 00 00 00 2D E3 5A>
<L[3]
 <U2[1] 3>
 <U4[1] 103>
 <L[1]
  <L[2]
   <U4[1] 103>
   <L[1]
    <A[23] "67.5x75-T-014-014-N0191">
   >
  >
 >
>
.
2022/3/7 11:26:32.854: Send S6F12 <Header: 00 00 06 0C 00 00 00 2D E3 5A>
<B[1] 0x00>
.
```

NGBOAT MOVEOUT LOG

Description:

NGBoatBarcode attached to the NGBoatMoveOutEvent(104).

```
2022/3/7 11:27:56.379: Receive S6F11 <Header: 00 00 86 0B 00 00 00 2D E3 5C>
<L[3]
 <U2[1] 4>
 <U4[1] 104>
 <L[1]
  <L[2]
   <U4[1] 104>
   <L[1]
    <A[23] "67.5x75-T-014-014-N0191">
   >
  >
 >
>
.
2022/3/7 11:27:56.382: Send S6F12 <Header: 00 00 06 0C 00 00 00 2D E3 5C>
<B[1] 0x00>
.
```

EMPTYBOAT RELEASE LOG

Description:

EmptyBoatBarcode attached to the EmptyBoatReleaseEvent(105).

```
2022/5/17 09:58:28.612: Receive S6F11 <Header: 00 00 86 0B 00 00 00 6B DB F8>
<L[3]
 <U2[1] 9>
 <U4[1] 105>
 <L[1]
  <L[2]
   <U4[1] 105>
   <L[1]
    <A[23] "67.5x75-T-014-014-N0191">
   >
  >
 >
>
.
2022/5/17 09:58:28.615: Send S6F12 <Header: 00 00 06 0C 00 00 00 6B DB F8>
<B[1] 0x00>
.
```

```
2022/5/17 09:58:30.099: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[17] "EMPTYBOAT_RELEASE">
 <L[2]
  <L[2]
   <A[17] "EMPTYBOAT_BARCODE">
   <A[23] "67.5x75-T-014-014-N0191">
  >
  <L[2]
   <A[24] "EMPTYBOAT_RELEASE_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/5/17 09:58:30.100: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with EMPTYBOAT_RELEASE_STATUS

PASS1MAGAZINE MOVEIN LOG

Description:

Pass1MagazineID attached to the Pass1MagazineMoveInEvent(106).

```
2022/3/7 11:31:36.131: Receive S6F11 <Header: 00 00 86 0B 00 00 00 2D E3 61>
<L[3]
 <U2[1] 6>
 <U4[1] 106>
 <L[1]
  <L[2]
   <U4[1] 106>
   <L[1]
    <A[9] "MA12965WE">
   >
  >
 >
>
.
2022/3/7 11:31:36.133: Send S6F12 <Header: 00 00 06 0C 00 00 00 2D E3 61>
<B[1] 0x00>
.
```

```
2022/3/7 11:38:57.635: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[20] "PASS1MAGAZINE_MOVEIN">
 <L[2]
  <L[2]
   <A[16] "PASS1MAGAZINE_ID">
   <A[9] "MA12965WE">
  >
  <L[2]
   <A[27] "PASS1MAGAZINE_MOVEIN_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/3/7 11:38:57.636: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with PASS1MAGAZINE_MOVEIN_STATUS

PASS2MAGAZINE MOVEIN LOG

Description:

Pass2MagazineID attached to the Pass2MagazineMoveInEvent(107).

```
2022/3/7 11:49:42.747: Receive S6F11 <Header: 00 00 86 0B 00 00 00 2D E3 72>
<L[3]
 <U2[1] 7>
 <U4[1] 107>
 <L[1]
  <L[2]
   <U4[1] 107>
   <L[1]
    <A[9] "MA12965WE">
   >
  >
 >
>
.
2022/3/7 11:49:42.751: Send S6F12 <Header: 00 00 06 0C 00 00 00 2D E3 72>
<B[1] 0x00>
.
```

```
2022/3/7 11:48:55.316: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[20] "PASS2MAGAZINE_MOVEIN">
 <L[2]
  <L[2]
   <A[16] "PASS2MAGAZINE_ID">
   <A[9] "MA12965WE">
  >
  <L[2]
   <A[27] "PASS2MAGAZINE_MOVEIN_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/3/7 11:48:55.317: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with PASS2MAGAZINE_MOVEIN_STATUS

NGMAGAZINE MOVEIN LOG

Description:

NGMagazineID attached to the NGMagazineMoveInEvent(108).

```
2022/3/7 11:51:11.195: Receive S6F11 <Header: 00 00 86 0B 00 00 00 2D E3 74>
<L[3]
 <U2[1] 8>
 <U4[1] 108>
 <L[1]
  <L[2]
   <U4[1] 108>
   <L[1]
    <A[9] "MA12965WE">
   >
  >
 >
>
.
2022/3/7 11:51:11.200: Send S6F12 <Header: 00 00 06 0C 00 00 00 2D E3 74>
<B[1] 0x00>
.
```

```
2022/3/7 11:54:15.899: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[17] "NGMAGAZINE_MOVEIN">
 <L[2]
  <L[2]
   <A[13] "NGMAGAZINE_ID">
   <A[9] "MA12965WE">
  >
  <L[2]
   <A[24] "NGMAGAZINE_MOVEIN_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/3/7 11:54:15.899: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with NGMAGAZINE_MOVEIN_STATUS

EMPTYMAGAZINE MOVEIN LOG

Description:

EmptyMagazineID attached to the EmptyMagazineMoveInEvent(109).

```
2022/3/7 11:58:58.468: Receive S6F11 <Header: 00 00 86 0B 00 00 00 2D E3 7C>
<L[3]
 <U2[1] 9>
 <U4[1] 109>
 <L[1]
  <L[2]
   <U4[1] 109>
   <L[1]
    <A[9] "MA12965WE">
   >
  >
 >
>
.
2022/3/7 11:58:58.472: Send S6F12 <Header: 00 00 06 0C 00 00 00 2D E3 7C>
<B[1] 0x00>
.
```

```
2022/3/7 13:16:40.232: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[20] "EMPTYMAGAZINE_MOVEIN">
 <L[2]
  <L[2]
   <A[16] "EMPTYMAGAZINE_ID">
   <A[9] "MA12965WE">
  >
  <L[2]
   <A[27] "EMPTYMAGAZINE_MOVEIN_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/3/7 13:16:40.232: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with EMPTYMAGAZINE_MOVEIN_STATUS

PASS1MAGAZINE MOVEOUT LOG

Description:

Pass1MagazineID attached to the Pass1MagazineMoveOutEvent(110).

```
2022/5/17 10:25:10.163: Receive S6F11 <Header: 00 00 86 0B 00 00 00 15 CD 22>
<L[3]
 <U2[1] 2>
 <U4[1] 110>
 <L[1]
  <L[2]
   <U4[1] 110>
   <L[3]
    <A[10] "SD58767868">
    <A[9] "MA12965WE">
    <A[3] "965">
   >
  >
 >
>
.
2022/5/17 10:25:10.166: Send S6F12 <Header: 00 00 06 0C 00 00 00 15 CD 22>
<B[1] 0x00>
.
```

```
2022/5/17 10:25:11.484: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[21] "PASS1MAGAZINE_MOVEOUT">
 <L[2]
  <L[2]
   <A[16] "PASS1MAGAZINE_ID">
   <A[9] "MA12965WE">
  >
  <L[2]
   <A[28] "PASS1MAGAZINE_MOVEOUT_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/5/17 10:25:11.489: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with PASS1MAGAZINE_MOVEOUT_STATUS

PASS2MAGAZINE MOVEOUT LOG

Description:

Pass2MagazineID attached to the Pass2MagazineMoveOutEvent(111).

```
2022/5/17 10:27:54.507: Receive S6F11 <Header: 00 00 86 0B 00 00 00 15 CD 25>
<L[3]
 <U2[1] 3>
 <U4[1] 111>
 <L[1]
  <L[2]
   <U4[1] 111>
   <L[3]
    <A[10] "SD58767868">
    <A[9] "MA12965WE">
    <A[3] "806">
   >
  >
 >
>
.
2022/5/17 10:27:54.510: Send S6F12 <Header: 00 00 06 0C 00 00 00 15 CD 25>
<B[1] 0x00>
.
```

```
2022/5/17 10:28:00.011: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[21] "PASS2MAGAZINE_MOVEOUT">
 <L[2]
  <L[2]
   <A[16] "PASS2MAGAZINE_ID">
   <A[9] "MA12965WE">
  >
  <L[2]
   <A[28] "PASS2MAGAZINE_MOVEOUT_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/5/17 10:28:00.011: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with PASS2MAGAZINE_MOVEOUT_STATUS

NGMAGAZINE MOVEOUT LOG

Description:

NGMagazineID attached to the NGMagazineMoveOutEvent(112).

```
2022/5/17 10:30:24.258: Receive S6F11 <Header: 00 00 86 0B 00 00 00 15 CD 28>
<L[3]
 <U2[1] 4>
 <U4[1] 112>
 <L[1]
  <L[2]
   <U4[1] 112>
   <L[3]
    <A[10] "SD58767868">
    <A[9] "MA12965WE">
    <A[3] "856">
   >
  >
 >
>
.
2022/5/17 10:30:24.261: Send S6F12 <Header: 00 00 06 0C 00 00 00 15 CD 28>
<B[1] 0x00>
.
```

```
2022/5/17 10:30:26.306: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[18] "NGMAGAZINE_MOVEOUT">
 <L[2]
  <L[2]
   <A[13] "NGMAGAZINE_ID">
   <A[9] "MA12965WE">
  >
  <L[2]
   <A[25] "NGMAGAZINE_MOVEOUT_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/5/17 10:30:26.307: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with NGMAGAZINE_MOVEOUT_STATUS

BOAT ASSING NGBOAT LOG

Description:

BoatAssingNGBoatBarcode attached to the BoatAssingNGBoatEvent(113).

```
2022/5/17 10:39:05.347: Receive S6F11 <Header: 00 00 86 0B 00 00 00 15 CD 31>
<L[3]
 <U2[1] 6>
 <U4[1] 113>
 <L[1]
  <L[2]
   <U4[1] 113>
   <L[2]
    <A[10] "SD58767868">
    <A[23] "67.5x75-T-014-014-N0191">
   >
  >
 >
>
.
2022/5/17 10:39:05.350: Send S6F12 <Header: 00 00 06 0C 00 00 00 15 CD 31>
<B[1] 0x00>
.
```

```
2022/5/17 10:39:07.587: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[18] "BOAT_ASSING_NGBOAT">
 <L[2]
  <L[2]
   <A[26] "BOAT_ASSING_NGBOAT_BARCODE">
   <A[23] "67.5x75-T-014-014-N0191">
  >
  <L[2]
   <A[25] "BOAT_ASSING_NGBOAT_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/5/17 10:39:07.587: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with BOAT_ASSING_NGBOAT_STATUS

UI LOCK COMMAND LOG

Description:

UI Lock state switched result attached to the RemoteActionResultEvent(114).

★Remote control UI lock log. (SUCCESS or FAIL)

```
2022/3/16 10:35:54.181: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[15] "UI_LOCK_COMMAND">
 <L[1]
  <L[2]
   <A[13] "UI_LOCK_STATE">
   <A[4] "LOCK">
  >
 >
>
.
2022/3/16 10:35:54.187: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
2022/3/16 10:36:53.233: Receive S6F11 <Header: 00 00 86 0B 00 00 00 00 BB 83>
<L[3]
 <U2[1] 1>
 <U4[1] 114>
 <L[1]
  <L[2]
   <U4[1] 114>
   <L[1]
    <A[7] "SUCCESS">
   >
  >
 >
>
.
2022/3/16 10:36:53.236: Send S6F12 <Header: 00 00 06 0C 00 00 00 00 BB 83>
<B[1] 0x00>
.
```

★Remote control UI unlock log. (SUCCESS or FAIL)

```
2022/3/16 10:44:31.421: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[15] "UI_LOCK_COMMAND">
 <L[1]
  <L[2]
   <A[13] "UI_LOCK_STATE">
   <A[6] "UNLOCK">
  >
 >
>
.
2022/3/16 10:44:31.422: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
2022/3/16 10:44:50.447: Receive S6F11 <Header: 00 00 86 0B 00 00 00 00 BB 8B>
<L[3]
 <U2[1] 2>
 <U4[1] 114>
 <L[1]
  <L[2]
   <U4[1] 114>
   <L[1]
    <A[4] "FAIL">
   >
  >
 >
>
.
2022/3/16 10:44:50.449: Send S6F12 <Header: 00 00 06 0C 00 00 00 00 BB 8B>
<B[1] 0x00>
.
```

EQUIPMENT OPERATIONAL COMMAND LOG

Description:

Equipment Operational state switched result attached to the RemoteActionResultEvent(114).

★Remote control equipment start log. (SUCCESS or FAIL)

```
2022/3/16 10:51:23.902: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[29] "EQUIPMENT_OPERATIONAL_COMMAND">
 <L[1]
  <L[2]
   <A[27] "EQUIPMENT_OPERATIONAL_STATE">
   <A[5] "START">
  >
 >
>

2022/3/16 10:51:23.903: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>

2022/3/16 10:51:41.775: Receive S6F11 <Header: 00 00 86 0B 00 00 00 00 BB 92>
<L[3]
 <U2[1] 3>
 <U4[1] 114>
 <L[1]
  <L[2]
   <U4[1] 114>
   <L[1]
    <A[7] "SUCCESS">
   >
  >
 >
>

2022/3/16 10:51:41.779: Send S6F12 <Header: 00 00 06 0C 00 00 00 00 BB 92>
<B[1] 0x00>
```

★Remote control equipment stop log. (SUCCESS or FAIL)

```
2022/3/16 10:54:08.518: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[29] "EQUIPMENT_OPERATIONAL_COMMAND">
 <L[1]
  <L[2]
   <A[27] "EQUIPMENT_OPERATIONAL_STATE">
   <A[4] "STOP">
  >
 >
>
.
2022/3/16 10:54:08.519: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
2022/3/16 10:54:12.206: Receive S6F11 <Header: 00 00 86 0B 00 00 00 00 BB 95>
<L[3]
 <U2[1] 4>
 <U4[1] 114>
 <L[1]
  <L[2]
   <U4[1] 114>
   <L[1]
    <A[7] "SUCCESS">
   >
  >
 >
>
.
2022/3/16 10:54:12.209: Send S6F12 <Header: 00 00 06 0C 00 00 00 00 BB 95>
<B[1] 0x00>
.
```

EQUIPMENT ENDLOT COMMAND LOG

Description:

Equipment endlot result attached to the RemoteActionResultEvent(114).

```
2022/3/16 10:57:25.847: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[24] "EQUIPMENT_ENDLOT_COMMAND">
 <L[1]
  <L[2]
   <A[16] "EQUIPMENT_ENDLOT">
   <A[6] "ENDLOT">
  >
 >
>
.
2022/3/16 10:57:25.847: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
2022/3/16 10:57:29.391: Receive S6F11 <Header: 00 00 86 0B 00 00 00 00 BB 99>
<L[3]
 <U2[1] 5>
 <U4[1] 114>
 <L[1]
  <L[2]
   <U4[1] 114>
   <L[1]
    <A[7] "SUCCESS">
   >
  >
 >
>
.
2022/3/16 10:57:29.394: Send S6F12 <Header: 00 00 06 0C 00 00 00 00 BB 99>
<B[1] 0x00>
.
```

EQUIPMENT STARTRUN CHECK LOG

Description:

RecipeName attached to the EquipmentStartRunEvent(115).

```
2022/5/17 10:51:40.883: Receive S6F11 <Header: 00 00 86 0B 00 00 00 15 CD 3F>
<L[3]
 <U2[1] 8>
 <U4[1] 115>
 <L[1]
  <L[2]
   <U4[1] 115>
   <L[1]
    <A[8] "Recipe_A">
   >
  >
 >
>
.
2022/5/17 10:51:40.886: Send S6F12 <Header: 00 00 06 0C 00 00 00 15 CD 3F>
<B[1] 0x00>
.
```

```
2022/5/17 10:51:41.883: Send S2F41 <Header: 00 00 02 29 00 00 00 00 00 00>
<L[2]
 <A[18] "EQUIPMENT_STARTRUN">
 <L[1]
  <L[2]
   <A[25] "EQUIPMENT_STARTRUN_STATUS">
   <A[4] "PASS">
  >
 >
>
.
2022/5/17 10:51:41.884: Receive S2F42 <Header: 00 00 02 2A 00 00 00 00 00 00>
<L[2]
 <B[1] 0x00>
 <L[0]
 >
>
.
```

※Please return “PASS” or “NG” with EQUIPMENT_STARTRUN_STATUS